MODULE

First Semester, AY 2019-2020

NETWORK 1, FUNDAMENTALS

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CERTIFICATE OF APPROVAL FOR UTILIZATION

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MODULE

NETWORKING I, FUNDAMENTALS

First Semester, AY 2019-2020

ROSALIE B. SISON
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UNIT 1

Networking - An Overview

At the end of the unit the student should be able to:

1. Define the network
2. Identify the basic networking concept and terms
3. Distinguish the different network
4. Determine the general network design concepts
5. Design a simple LAN

What is a Network?

A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

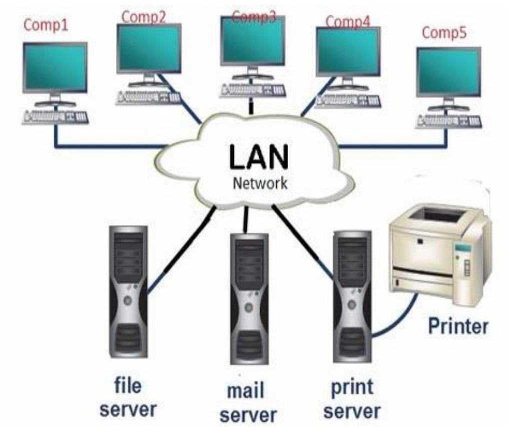
Types of Networks

1Personal Area Network (PAN)

The smallest and most basic type of network, a PAN is made up of a wireless modem, a computer or two, phones, printers, tablets, etc., and revolves around one person in one building. These types of networks are typically found in small offices or residences, and are managed by one person or organization from a single device.

2. Local Area Network (LAN)

We're confident that you've heard of these types of networks before – LANs are the most frequently discussed networks, one of the most common, one of the most original and one of the simplest types of networks. LANs connect groups of computers and low-voltage devices together across short distances (within a building or between a group of two or three buildings in close proximity to each other) to share information and resources. Enterprises typically manage and maintain LANs. Using routers, LANs can connect to wide area networks (WANs, explained below) to rapidly and safely transfer data.

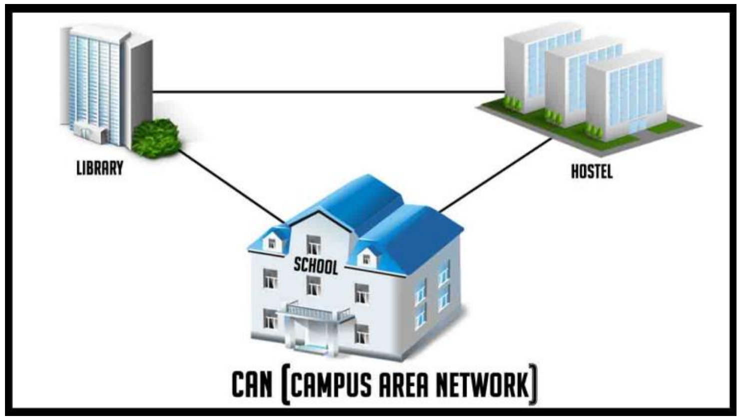


3. **Wireless Local Area Network (WLAN)**

Functioning like a LAN, WLANs make use of wireless network technology, such as Wi-Fi. Typically seen in the same types of applications as LANs, these types of networks don't require that devices rely on physical cables to connect to the network.

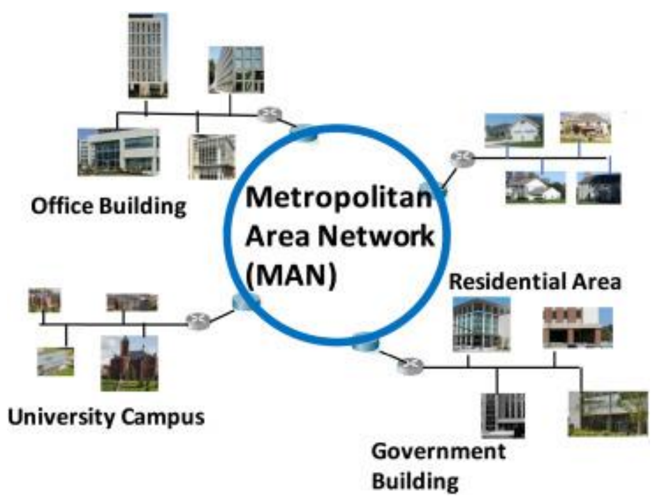
4. **Campus Area Network (CAN)**

Larger than LANs, but smaller than metropolitan area networks (MANs, explained below), these types of networks are typically seen in universities, large K-12 school districts or small businesses. They can be spread across several buildings that are fairly close to each other so users can share resources.



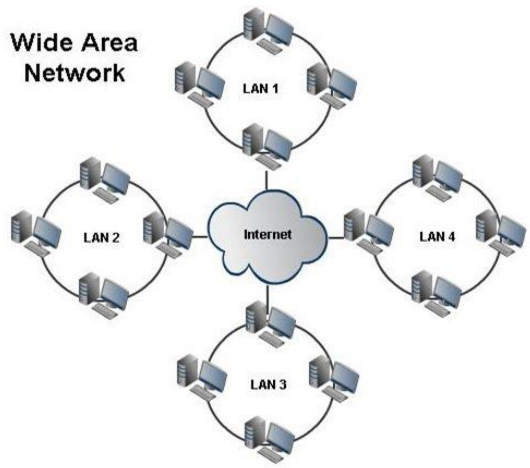
5. **Metropolitan Area Network (MAN)**

These types of networks are larger than LANs but smaller than WANs and incorporate elements from both types of networks. MANs span an entire geographic area (typically a town or city, but sometimes a campus). Ownership and maintenance are handled by either a single person or company (a local council, a large company, etc.)



6. Wide Area Network (WAN)

Slightly more complex than a LAN, a WAN connects computers together across longer physical distances. This allows computers and low-voltage devices to be remotely connected to each other over one large network to communicate even when they're miles apart. The Internet is the most basic example of a WAN, connecting all computers together around the world. Because of a WAN's vast reach, it is typically owned and maintained by multiple administrators or the public.



Boundaries between network

Boundary networks are where computers that are not NAP compliant can access a remediation server and become compliant. Once compliant, they can access an HCS Server and acquire a health certificate to participate in the secure network. Computers on the boundary network will accept communication requests from computers with a health certificate or without—this is how remediation occurs. Both the restricted network and the secure network have access to the boundary network.

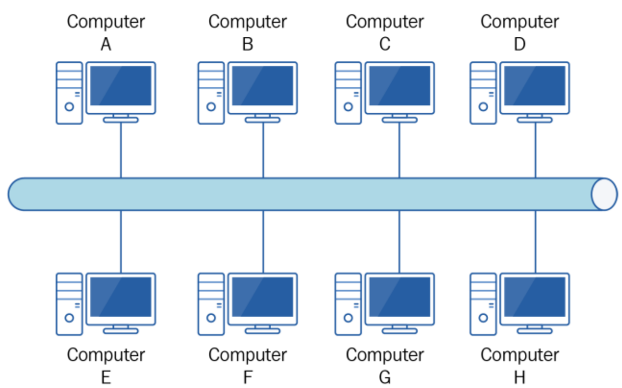
What is Network Topology?

Network topology refers to the manner in which the links and nodes of a network are arranged to relate to each other. Topologies are categorized as either physical network topology, which is the physical signal transmission medium, or logical network topology, which refers to the manner in which data travels through the network between devices, independent of physical connection of the devices. Logical network topology examples include twisted pair Ethernet, which is categorized as a logical bus topology, and token ring, which is categorized as a logical ring topology.

Physical network topology examples include star, mesh, tree, ring, point-to-point, circular, hybrid, and bus topology networks, each consisting of different configurations of nodes and links. The ideal network topology depends on each business's size, scale, goals, and budget. A network topology diagram helps visualize the communicating devices, which are modeled as nodes, and the connections between the devices, which are modeled as links between the nodes.

Types of Network Topology

Bus topology, also known as line topology, is a type of network topology in which all devices in the network are connected by one central RJ-45 network cable or coaxial cable. The single cable, where all data is transmitted between devices, is referred to as the bus, backbone, or trunk.



There are two types of bus topologies:

Linear bus topology. All devices are connected to a single cable with two end points.

Distributed bus topology. All devices are connected to a single cable that branches off into multiple sections, resulting in more than two end points.

Advantages of Bus Topology

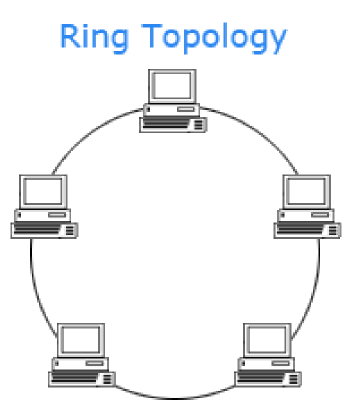
- Works efficiently for small networks
- Easy and cost-effective to install and add or remove devices
- Doesn't require as much cabling as alternative topologies
- If one device fails, other devices are not impacted

Disadvantages of Bus Topology

- If the cable is damaged, the entire network will fail or be split
- Difficult to troubleshoot problems
- Very slow and not ideal for larger networks
- Requires terminators at both ends of the cable to prevent bouncing signals that cause interference
- Adding more devices and more network traffic decreases the entire network's performance
- Low security due to all devices receiving the same signal from the source

RING TOPOLOGY

A **ring topology** is a network configuration where device connections create a circular data path. Each networked device is connected to two others, like points on a circle. Together, devices in a ring topology are referred to as a **ring network**. In a ring network, packets of data travel from one device to the next until they reach their destination. Most ring topologies allow packets to travel only in one direction, called a **unidirectional** ring network. Others permit data to move in either direction, called **bidirectional**. The major disadvantage of a ring topology is that if any individual connection in the ring is broken, the entire network is affected. Ring topologies may be used in either LANs (local area networks) or WANs (wide area networks). Depending on the network card used in each computer of the ring topology, a coaxial cable or an RJ-45 network cable is used to connect computers together.



Advantages of a ring topology

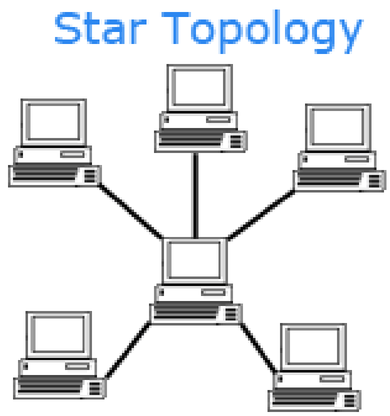
- All data flows in one direction, reducing the chance of packet collisions.
- A network server is not needed to control network connectivity between each workstation.
- Data can transfer between workstations at high speeds.
- Additional workstations can be added without impacting performance of the network.

Disadvantages of a ring topology

- All data being transferred over the network must pass through each workstation on the network, which can make it slower than a star topology.
- The entire network will be impacted if one workstation shuts down.
- The hardware needed to connect each workstation to the network is more expensive than Ethernet cards and hubs/switches.

STAR TOPOLOGY

Alternatively referred to as a **star network**, **star topology** is one of the most common network setups. In this configuration, every node connects to a central network device, like a hub, switch, or computer. The central network device acts as a server and the peripheral devices act as clients. In a star topology setup, either a coaxial or RJ-45 network cable is used, depending on the type of network card installed in each computer. The image shows how this network setup gets its name, as it is shaped like a star.



Advantages of star topology

- Centralized management of the network, through the use of the central computer, hub, or switch.
- Easy to add another computer to the network.
- If one computer on the network fails, the rest of the network continues to function normally.

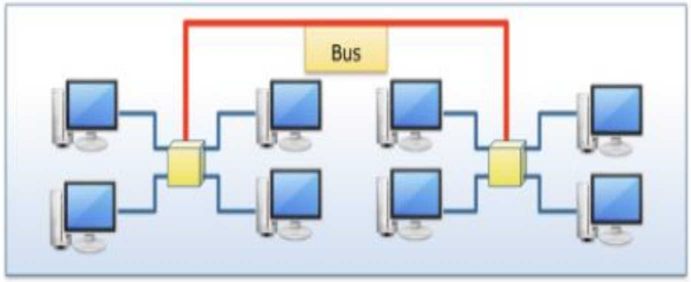
Disadvantages of star topology

- May have a higher cost to implement, especially when using a switch or router as the central network device.
- The central network device determines the performance and number of nodes the network can handle.
- If the central computer, hub, or switch fails, the entire network goes down and all computers are disconnected from the network.

Star/Bus hybrid topology

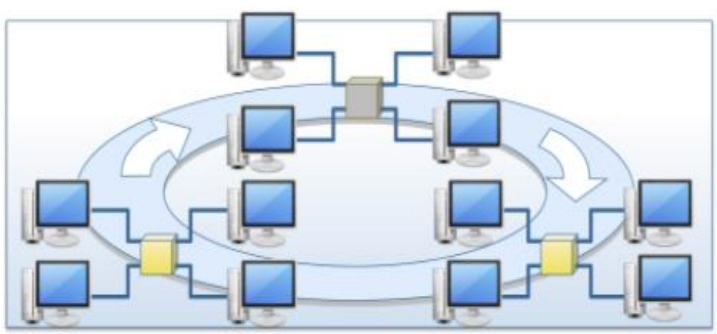
In a star/bus topology, several star topology networks are linked to a bus connection. When the star configuration is busy, you can add a second star configuration and connect the two-star topologies using a bus connection.

In a star/bus topology, if a single computer fails, the rest of the network is not affected. However, if the central component, the so-called hub, that connects all computers in a star topology to each other, fails, all computers connected to that component fail and can no longer communicate.



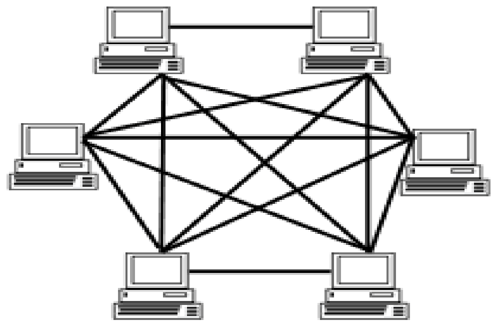
Star/Ring hybrid topology

In the star/ring topology, the computers are connected to a central component as in a star network. However, these components are wired in the form of a ring network. If a single computer fails, the rest of the network is not affected, as with the star/bus topology. Using token passing, every computer in a star/ring topology has the same communication opportunity. This allows a higher network traffic between segments than with the star/bus topology.



Mesh Topology

A mesh topology is a network setup where each computer and network device is interconnected with one another. This topology setup allows for most transmissions to be distributed even if one of the connections goes down. It is a topology commonly used for wireless networks. Below is a visual example of a simple computer setup on a network using a **mesh topology**.



Different types of mesh topology

- There are two forms of this topology: full mesh and a partially-connected mesh.
- In a *full mesh topology*, every computer in the network has a connection to each of the other computers in that network. The number of connections in this network can be calculated using the following formula (*n* is the number of computers in the network): n(n-1)/2
- In a *partially-connected mesh topology*, at least two of the computers in the network have connections to multiple other computers in that network. It is an inexpensive way to implement redundancy in a network. If one of the primary computers or connections in the network fails, the rest of the network continues to operate normally.

Advantages of a mesh topology

- Manages high amounts of traffic, because multiple devices can transmit data simultaneously.
- A failure of one device does not cause a break in the network or transmission of data.
- Adding additional devices does not disrupt data transmission between other devices.

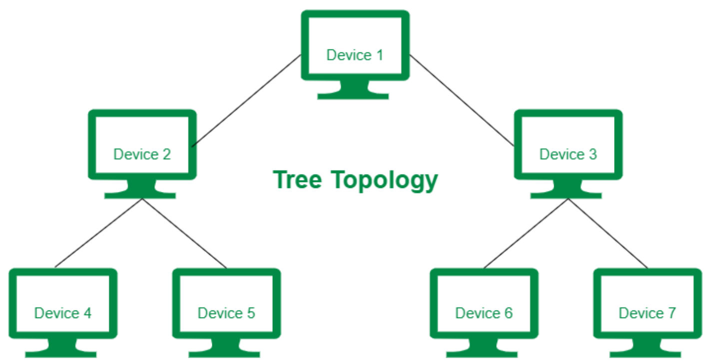
Disadvantages of a mesh topology

- The cost to implement is higher than other network topologies, making it a less desirable option.
- Building and maintaining the topology is difficult and time consuming.

- The chance of redundant connections is high, which adds to the high costs and potential for reduced efficiency.

Tree Topology

Tree Topology is a topology which is having a tree structure in which all the computer are connected like the branches which are connected with the tree. In Computer Network, tree topology is called as a combination of a Bus and Start network topology. The main advantages of this topology are these are very flexible and also have better scalability. Tree network topology is considered to be the simplest topology in all the topologies which is having only one route between any two nodes on the network. The pattern of connection resembles a tree in which all branches spring from one root hence (Tree Topology). Tree topology is one of the most popular among five network topologies.



Advantages of Tree Topology:

- This topology is the combination of bus and star topology.
- This topology provides a hierarchical as well as central data arrangement of the nodes.
- As the leaf nodes can add one or more nodes in the hierarchical chain, this topology provides high scalability.
- The other nodes in a network are not affected, if one of their nodes get damaged or not working.
- Tree topology provides easy maintenance and easy fault identification can be done.
- A callable topology. Leaf nodes can hold more nodes.
- Supported by several hardware and software vendors.

- Point-to-point wiring for individual segments.

Disadvantages of Tree Topology:

- This network is very difficult to configure as compared to the other network topologies.
- Length of a segment is limited & the limit of the segment depends on the type of cabling used.
- Due to the presence of large number of nodes, the network performance of tree topology becomes a bit slowly.
- If the computer in first level is erroneous, next level computer will also go under problems.
- Requires large number of cables compared to star and ring topology.
- As the data needs to travel from the central cable this creates dense network traffic.
- The Backbone appears as the failure point of the entire segment of the network.
- Treatment of the topology is pretty complex.
- The establishment cost increases as well.
- If the bulk of nodes are added in this network, then the maintenance will become complicated.

Network Design

Network design is a category of systems design that deals with data transport mechanisms. As with other systems' design disciplines, network design follows an analysis stage, where requirements are generated, and precedes implementation, where the system (or relevant system component) is constructed. The objective of network design is to satisfy data communication requirements while minimizing expense. Requirement scope can vary widely from one network design project to another based on geographic particularities and the nature of the data requiring transport.

REVIEW QUESTIONS

Name:____**Score:**________________
Section: _______________________________________ **Date :** ________________

I. Write your answers on the space provide after the number.

_______1. A computer device manufacturer owns four buildings in downtown Raleigh, North Carolina, and the LAN in each building are connected together for integrated communications and resource sharing. This type of network is an example of which of the following?

A. LAN-NET

B. TAN

C. CAN

D. WAN

E. None of the above

_______2. Which of the following are ways to share a printer among computer users? (Choose all that apply.)

A. Create a joint area network.

B. Connect the printer to a Windows 7 workstation and share it on a network;

C. Purchase a printer that has a built-in NIC

D. Use a printer server device that connects to a network.

_______3. A bus topology must have which of the following at each end of the bus? (Choose all that apply.)

A. a router junction

B. a terminator

C. a capacitor

D. an amplification connector

_______4. What advice might a new network designer receive when just starting out? (Choose all that apply.)

A. Learn how networks work in terms of protocols, access methods, and topology; and d. Understand the physical equipment used in LANs and WANs.

B. Avoid the risks of implementing fault tolerance.

C. Start with the safe mesh design in which information is continuously sent around the network ensuring that if a computer misses it the first time ere are many other chances to obtain the same information without data loss.

D. Understand the physical equipment used in LANs and WANs.

_______5. Which of the following are reasons why you might have a network in your home of two adults and three teenagers? (Choose all that apply.)

A. accessing the Internet

B. sharing a printer

C. accessing entertainment resources, such as streaming movies

D. sharing files between computers

_______6. In your new building, the contractor is including secure wiring closets, cable, and cable pathways installed in the building as well as backbone cabling into a specialized server room. All of this cabling is part of the __________.

A. network data zone

B. cable plant

C. traffic zone

D. throughput byway

_______7. A large company has many computer resources including computers, servers, mainframes, network devices, computer development labs, computer databases, shared disk arrays, and so on. This is an example of which of the following?

A. an enterprise network

B. an extended client network

C. a radiating network

D. network concentration

_______8. Which of the following are the disadvantages of the traditional bus topology? (Choose all that apply.)

A. It cannot be connected to the Internet

B. It allows only for a maximum of 16 computers per bus

C. It is subject to network congestion, requiring additional network devices to control traffic flow;

D. One defective node or cable segment can take down a network.

_______9. Assume you are a network consultant for a company that is designing a private WAN to communicate between five locations spread throughout a city. You want to tell the company president that this WAN will use a design for maximum uptime to all locations. Which of the following designs should you use?

A. bus

B. star-ring hybrid

C. mesh

D. traditional ring

_______10. Assume you have been hired to design a network for a bank. When you determine the factors that affect the network design, you should look at which of the following? (Choose all that apply.)

A. software applications to be used

B. work patterns in the bank

C. the computers and operating systems to be connected to the network

D. the security required by the bank and bank regulators

_______11. Which network topology is no longer used much in modern LANs?

A. star

B. star-bus hybrid

C. ring

D. all of the above are commonly used in modern LANs.

_______12. Which of the following are examples of network nodes? (Choose all that apply.)

A. a computer attached to a network; and d. a network switch

B. a DVD/CD-ROM array attached to a server

C. a printer attached to a network computer

D. a network switch

_______13. When you design a network that has a bus or star-bus hybrid design, what organization's specifications should you check to make sure you follow the bus segment length requirements?

A. Microsoft

B. Red Hat

C. Institute of Electrical and Electronics Engineers

D. National Electrica Contractors Association

_______14. Your organization has a network that uses the star-bus hybrid topology. A new network associate is in the process of ordering spare network parts and has included 20 terminators in the order. Which of the following is your advice?

A. Increase the order terminators burn out quickly on busy star-bus networks

B. Be certain to order terminators that contain LEDs to ensure communications can go only one way on the network

C. Order200 ohm terminators, because the network contains over 50 connections requiring a high overall resistance.

D. Omit the portion of the order for terminators, because terminators are built into the devices.

_______15. High-speed WANs that use the ring topology may use which of the following?

A. two loops for redundant data transmission

B. ring connection that work like terminators

C. baseband optics

D. overflow NICs to regulate the speed of transmission

_______16. You are researching the possible benefits of a LAN for an advertising firm. Which of the following are the benefits that you might include in your report to the firm's partners? (Choose all that apply.)

A. A LAN would enable the firm to use a DVD/CD-ROM array for sharing commonly accessed DVDs and CDs.

B. A LAN would eliminate the need for laptop computers (plus laptop computers are not compatible with LAN communications)

C. Printer expenditures could be reduced; and d. Centralized backups of client files would be possible.

D. Centralized backups of client files would be possible.

_______17. Your company in New York City wants to open a branch office in Richmond, Virginia, and connect the branch office through the Internet. What type of network should be considered for enhanced security?

A. a closed transmission network

B. a VPN

C. a top-down security network

D. a star-secure network

E. none of the above

_______18. Which of the following devices might be used to connect computers in a star topology? (Choose all that apply.)

A. conjunction box

B. switch

C. terminator

D. router

_______19. You are setting up a network for your small business of four employees, each of whom has a computer and needs to periodically share files with other employees. Instead of having a server, the files are shared through each computer's operating system. This is an example of a(n) __________ network.

A. equality

B. peer-to-peer

C. jelly-based

D. open

_______20. ICS in Windows 7 and Windows Server 2008 R2 can be configured through which of the following?

A. Network Control icon in the taskbar

B. Network and Sharing Center

C. Wide Network Configuration option form the Start button

D. Office Features menu

UNIT 2

How LAN and WAN communication Work

At the end of the unit the student should be able to:

1. Recognize the different layers of OSI model, which sets standards for LAN and WAN communication
2. Differentiate the LAN and WAN communications.
3. Describe the major LAN transmission methods
4. Explain the basic WAN network communications topologies and transmission methods, including telecommunication, cable TV, satellite, and wireless technology
5. Explain the advantages of using Ethernet in network design

The OSI Model

The OSI Model (Open Systems Interconnection Model) is a conceptual framework used to describe the functions of a networking system. The OSI model characterizes computing functions into a universal set of rules and requirements in order to support interoperability between different products and software. In the OSI reference model, the communications between a computing system are split into seven different abstraction layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Created at a time when network computing was in its infancy, the OSI was published in 1984 by the International Organization for Standardization (ISO). Though it does not always map directly to specific systems, the OSI Model is still used today as a means to describe Network Architecture.

How the OSI model works

Information technology (IT) networking professionals use OSI to model or conceptualize how data is sent or received over a network. Understanding this is a foundational part of most IT networking certifications, including the Cisco Certified Network Associate (CCNA) and CompTIA Network+ certification programs. As mentioned, the model is designed to break down data transmission standards, processes and protocols over a series of seven layers, each of which is responsible for performing specific tasks concerning sending and receiving data. The main concept of OSI is that the process of communication between two endpoints in a network can be divided into seven distinct groups of related functions, or layers.

Each communicating user or program is on a device that can provide those seven layers of function. In this architecture, each layer serves the layer above it and, in turn, is served by the layer below it. So, in a given message between users, there will be a flow of data down through the layers in the source computer, across the network and then up through the layers in the receiving computer. Only the application layer at the top of the stack doesn't provide services to a higher-level layer. The seven layers of function are provided by a combination of applications, operating systems (OSes), network card device drivers, networking hardware and protocols that enable a system to transmit a signal over a network through various physical mediums, including twisted-pair copper, fiber optics, Wi-Fi or Long-Term Evolution (LTE) with 5G.

Physical Layer

Physical Layer is the bottom-most layer in the **Open System Interconnection (OSI) Model** which is a physical and electrical representation of the system. It consists of various network components such as power plugs, connectors, receivers, cable types, etc. Physical Layer sends data bits from one device(s) (like a computer) to another device(s). Physical Layer defines the types of encoding (that is how the 0's and 1's is encoded in a signal). Physical Layer is responsible for the communication of the unstructured raw data streams over a physical medium.

Functions Performed by Physical Layer:
Following are some important and basic functions that are performed by the Physical Layer of the OSI Model –

1. Physical Layer maintains the data rate (how many bits a sender can send per second).

2. It performs Synchronization of bits.

3. It helps in Transmission Medium decision (direction of data transfer).

4. It helps in Physical Topology (Mesh, Star, Bus, Ring) decision (Topology through which we can connect the devices with each other).

5. It helps in providing Physical Medium and Interface decisions.

6. It provides two types of configuration Point to Point configuration and Multi-Point configuration.

7. It provides an interface between devices (like PC's or computers) and transmission medium.

8. It has a protocol data unit in bits.

9. Hubs, Ethernet, etc. device is used in this layer.

10. This layer comes under the category of Hardware Layers (since the hardware layer is responsible for all the physical connection establishment and processing too).

11. It provides an important aspect called Modulation, which is the process of converting the data into radio waves by adding the information to an electrical or optical nerve signal.

12. It also provides Switching mechanism wherein data packets can be forward from one port (sender port) to the leading destination port.

Data link layer

Data Link Layer is second layer of OSI Layered Model. This layer is one of the most complicated layers and has complex functionalities and liabilities. Data link layer hides the details of underlying hardware and represents itself to upper layer as the medium to communicate. Data link layer works between two hosts which are directly connected in some sense. This direct connection could be point to point or broadcast. Systems on broadcast network are said to be on same link. The work of data link layer tends to get more complex when it is dealing with multiple hosts on single collision domain. Data link layer is responsible for converting data stream to signals bit by bit and to send that over the underlying hardware. At the receiving end, Data link layer picks up data from hardware which are in the form of electrical signals, assembles them in a recognizable frame format, and hands over to upper layer.

Data link layer has two sub-layers:

Logical Link Control: It deals with protocols, flow-control, and error control

Media Access Control: It deals with actual control of media

Functionality of Data-link Layer

Framing- Data-link layer takes packets from Network Layer and encapsulates them into Frames. Then, it sends each frame bit-by-bit on the hardware. At receiver' end, data link layer picks up signals from hardware and assembles them into frames.

Addressing-Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.

Synchronization-When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.

Error Control-Sometimes signals may have encountered problem in transition and the bits are flipped. These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.

Flow Control-Stations on same link may have different speed or capacity. Data-link layer ensures flow control that enables both machine to exchange data on same speed.

Multi-Access-When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple Systems.

NETWORK LAYER

The network layer is a portion of online communications that allows for the connection and transfer of data packets between different devices or networks.

The network layer is the third level (Layer 3) of the Open Systems Interconnection Model (OSI Model) and the layer that provides data routing paths for network communication. Data is transferred to the receiving device in the form of packets via logical network paths in an ordered format controlled by the network layer. Logical connection setup, data forwarding, routing and delivery error reporting are the network layer's primary responsibilities. Layer 3 can be either able to support connection-oriented or connectionless networks (but not both of them at the same time). The network layer is considered the backbone of the OSI Model. It selects and manages the best logical path (virtual circuit) for data transfer between nodes by assigning destination and source IP addresses to each data segment. In the OSI model, the network layer responds to requests from the layer above it (transport layer) and issues requests to the layer below it (data link layer). Information about the source and destination hosts is already contained in the data link layer (Layer 2), but it's the network layer that interconnects different networks allowing data to move between them. Quality of Service (QoS) allows for the prioritization of certain types of traffic. Data is checked for errors by the Internet Control Message Protocol (ICMP), which ensures that packets are sent correctly by Layer 3. Layer 3 contains hardware devices such as routers, bridges, firewalls and switches, but it actually creates a logical image of the most efficient communication route and implements it with a physical medium. Network layer protocols exist in every host or router. The router examines the header fields of all the IP packets that pass through it. Networking software is used to attach the header to each data packet sent as well as to read it to determine how the packet is handled at the receiving end. All the connection, encryption, checking, and routing processes occur at the network layer and are made by several different protocols, including:

- Internet Protocol (IP) and Netware IPX/SPX are the most common protocols associated with the network layer.
- The network layer infrastructure is inherently vulnerable to malicious attacks since it is exposed on the Internet. Distributed denial-of-service (DDoS) attacks can be launched to overwhelm all the physical network interfaces such as routers and stop data transmission.
- Although this comparison can be misleading, the OSI network layer is often referenced as the equivalent of the Internet layer of the TCP/IP model. However, there are several differences between the two, and the TCP/IP Internet layer only has a limited amount of the functions covered by the OSI network layer.

Transport Layer

The transport layer is the fourth layer in the open systems interconnection (OSI) network model. The OSI model divides the tasks involved with moving information between networked computers into seven smaller, more manageable task groups. Each of the seven OSI layers is assigned a task or group of tasks. The transport layer's tasks include error correction as well as segmenting and DE segmenting data before and after it's transported across the network. This layer is also responsible for flow control and making sure that segmented data is delivered over the network in the correct sequence. Layer 4 (the transport layer) uses the transmission control protocol (TCP) & user data protocol (UDP) to carry out its tasks.

What Services Can the Transport Layer Provide?

Connection-Oriented Communication

Devices at the end-points of a network communication establish a handshake protocol such as TCP to ensure a connection is robust before data is exchanged. The weakness of this method is that for each delivered message, there is a requirement for an acknowledgment, adding considerable network load compared to self-error-correcting packets. The repeated requests cause significant slowdown of network speed when defective byte streams or datagrams are sent.

Same Order Delivery

Ensures that packets are always delivered in strict sequence by assigning them a number. Although the network layer is responsible, the transport layer can fix any discrepancies in sequence caused by packet drops or device interruption by reordering them.

Data Integrity

Using checksums, the data integrity across all the delivery layers can be ensured. These checksums guarantee that the data transmitted is the same as the data received and that is not corrupt. Missing or corrupted data can be resent by requesting retransmission from other layers.

Flow Control

Devices at each end of a network connection often have no way of knowing each other's capabilities in terms of data throughput. Data can end up being sent faster than the speed at which the receiving device is able to buffer or process it. When this happens, buffer overruns can cause complete communication breakdowns. Conversely, if the receiving device is not receiving data fast enough, this causes a buffer underrun, which may well cause an unnecessary reduction in network performance. Flow control ensures that the data is sent at a rate that is acceptable for both sides by managing data flow.

Traffic Control

Digital communications networks are subject to bandwidth and processing speed restrictions, which can mean a huge amount of potential for data congestion on the network. This network congestion can affect almost every part of a network. The transport layer can identify the symptoms of overloaded nodes and reduced flow rates and take the proper steps to remediate these issues.

Multiplexing

The transmission of multiple packet streams from unrelated applications or other sources (multiplexing) across a network requires some very dedicated control mechanisms, which are found in the transport layer. This multiplexing allows the use of simultaneous applications over a network such as when different internet browsers are opened on the same computer.

Session Layer

The Session Layer is the 5th layer in the Open System Interconnection (OSI) model. This layer allows users on different machines to establish active communications sessions between them. It is responsible for establishing, maintaining, synchronizing, terminating sessions between end-user applications. In Session Layer, streams of data are received and further marked, which is then resynchronized properly, so that the ends of the messages are not cut initially and further data loss is avoided. This layer basically establishes a connection between the session entities. This layer handles and manipulates data which it receives from the Session Layer as well as from the Presentation Layer.

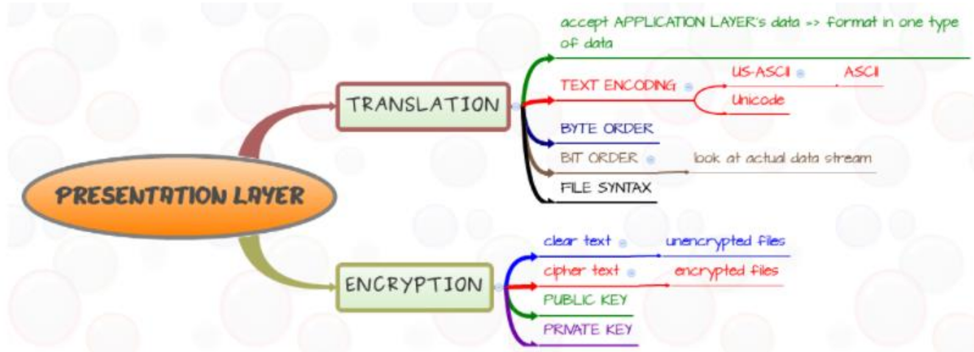
Following are some of the functions which are performed by Session Layer

- Session Layer works as a dialog controller through which it allows systems to communicate in either half-duplex mode or full duplex mode of communication.
- This layer is also responsible for token management, through which it prevents two users to simultaneously access or attempting the same critical operation.

- This layer allows synchronization by allowing the process of adding checkpoints, which are considered as synchronization points to the streams of data.
- This layer is also responsible for session checkpointing and recovery.
- This layer basically provides a mechanism of opening, closing and managing a session between the end-user application processes.
- The services offered by Session Layer are generally implemented in application environments using remote procedure calls (RPCs).
- The Session Layer is also responsible for synchronizing information from different sources.
- This layer also controls single or multiple connections for each-end user application and directly communicates with both Presentation and transport layers.
- Session Layer creates procedures for checkpointing followed by adjournment, restart and termination.
- Session Layer uses checkpoints to enable communication sessions which are to be resumed from that particular checkpoint at which communication failure has occurred.
- The session Layer is responsible for fetching or receiving data information from its previous layer (transport layer) and further sends data to the layer after it (presentation layer).

What is presentation layer?

The presentation layer is located at the sixth level of the OSI model, it is responsible for the delivery and formatting of information to the application layer for further processing or display. This type of service is needed because different computer architectures use different data representations. In contrast to providing transparent data transport at the fifth level, the presentation layer handles all issues related to data presentation and transport, including translation, encryption, and compression.



Presentation layer functions

1. Network security and confidentiality management, text compression and packaging, virtual terminal protocol (VTP).

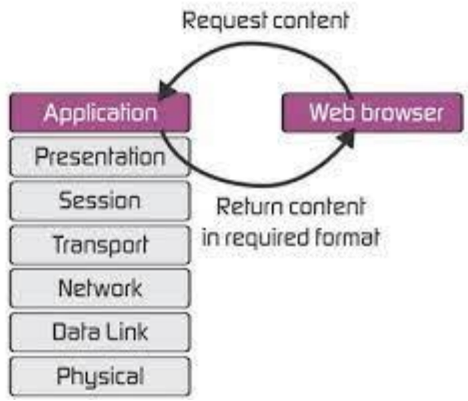
2. Syntax conversion - The abstract syntax is converted to the transfer syntax, and the other side to achieve the opposite conversion (transfer syntax will be converted to abstract syntax). Involved in the contents of the code conversion, character conversion, data format modification, as well as data structure operation adaptation, data compression, encryption and so on.

3. Grammar negotiation - According to the requirements of the application layer to negotiate the appropriate choice of context, that is, to determine the transmission syntax and transmission.

4. Connection management - Including the use of the session layer service to establish a connection, manage data transport and synchronization control over this connection (using the corresponding services at the session level), and terminate the connection either normally or absently.

What is application layer?

The Application Layer is the seventh layer of the seven-layer OSI model. Application layer interface directly interacts with the application and provides common web application services. The application layer also makes a request to the presentation layer. Application layer is the highest level of open systems, providing services directly for the application process.



Application layer functions

- Transport access and management
- It allows a user to access, retrieve and manage files in a remote computer.
- mail services
- It provides the basis for email forwarding and storage facilities.
- Virtual terminal
- For various reasons, it can be said that the standardization of terminals has completely failed. The OSI solution to this problem is to define a virtual terminal that is really just an abstract data structure that takes the abstract state of the actual terminal. This abstract data structure can be operated by both the keyboard and the computer and reflects the current state of the data structure on the display. The computer can query this abstract data structure and change this abstract data structure so that the output appears on the screen.
- **Other functions**
- In addition to the three functions above, there are some other functions: directory services, remote job entry, graphics, information communication and so on.

Application layer examples

- DNS (Domain Name System)
- The Domain Name System (DNS) is a hierarchical decentralized naming system for computers, services, or other resources connected to the Internet or a private network. It associates various

information with domain names assigned to each of the participating entities.

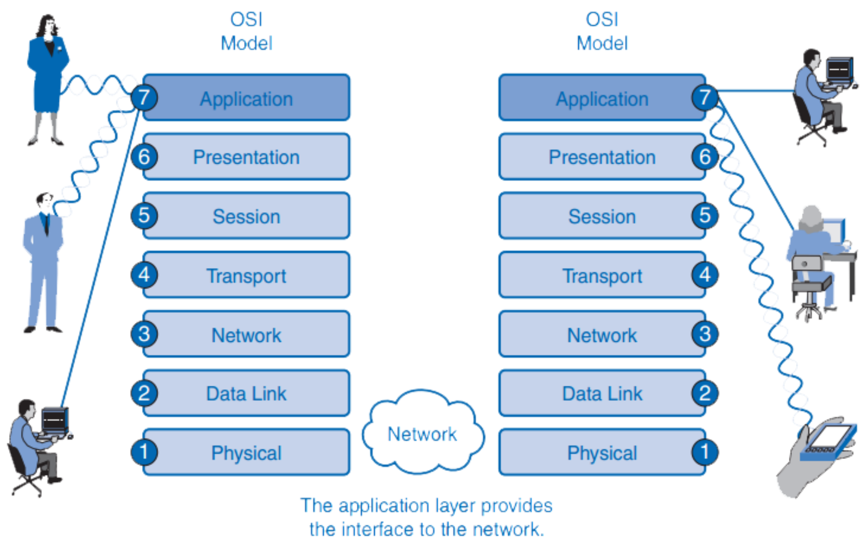
- Currently, the limit on domain name length is 63 characters, including www. And .com or other extensions. Domain names are also restricted to only a subset of ASCII characters, making many other languages unable to properly represent their names and words. Punycode-based IDNA systems, which map Unicode strings to valid DNS character sets, have been validated and adopted by some registries as a workaround.
- HTTP (Hypertext Transfer Protocol)
- The Hypertext Transfer Protocol (HTTP) is an application protocol for distributed, collaborative, and hypermedia information systems.[1] HTTP is the foundation of data communication for the World Wide Web.
- HTTP is a client and server-side standard for request and response (TCP). The client is the end user, the server is the website. Using a web browser, web crawler, or other tools, the client initiates an HTTP request to the specified port on the server. The responding server stores (some) resources, such as HTML files and images. (We call it)
- Typically, a request is made by the HTTP client to establish a TCP connection to the server's designated port (the default is port 80). The HTTP server listens on that port for requests sent by the client. Upon receipt of the request, the server sends a status line (to the client), such as "HTTP / 1.1 200 OK," and (response) a message body that may be a requested file, an error message, or some other information.

FTP (File Transfer Protocol)

FTP services generally run on both ports 20 and 21. Port 20 is used to transmit data flow between the client and server, while port 21 is used to transmit control flow and is the command to import to the ftp server. When data is streamed, the control flow is idle. When the control flow, idle for a long time, the client's firewall, the session will be set to overtime, so when a large amount of data through the firewall, there will be some problems. At this point, although the file can be successfully transmitted, it can be broken by the firewall because of the control session; the transfer can result in some errors.

Communication between Stacks

When a message is sent from one machine to another, it travels down the layers on one machine and then up the layers on the other machine,



ACTIVITY ANALYSIS

In the following assignments, you consult for the newsroom of the Franklin Daily Herald, a newspaper in a Midwestern U.S. city that is still thriving because of its close ties to the community and first-rate reporting. All of the news reporters have laptop computers, and now their management has funded a project to network the entire newspaper building. Having a network that links all of the departments and enables company-wide exchange of information and Internet access will help position this newspaper to stay competitive with other news sources.

Activity Analysis 1: Expanding the Network Backbone

There is already a partial network backbone using Ethernet in the newspaper's building that joins some of the administrative and advertising offices. Prepare a report or slide show that explains why Ethernet is still a good choice for the network backbone.

Answer:

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Activity Analysis 2: Creating a Network That Can Communicate with Other Networks

The management wants to know if there are any guarantees that the network you are proposing will communicate with other networks. What is your response?

Answer:

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Activity Analysis 3: Questions About the OSI Model

Brett Mason, a new colleague with whom you are working at Network Design Consultants, is unsure about some aspects of the OSI model. He has a list of questions for you and asks that you develop a table that he can use as a reference for the answers. Create a table containing two columns and seven rows. Label the left column "Network Function," and label the right column "OSI Layer." Enter each of the following functions in its own row under the left column, and then specify the OSI layer that performs that function under the right column. Brett's questions about functions are as follows:

- Which layer resizes frames to match the receiving network?
- Which layer performs data compression?
- Which layer ensures data is received in the order it was sent?
- Which layer handles the data-carrying signal?
- Which layer provides file transfer services?
- Which layer enables routing?
- Which layer enables the receiving node to send an acknowledgment?

Answer:

UNIT 3

Using Network Communication Protocols

At the end of the unit the student should be able to:

1. Define the network protocols
2. Identify Legacy Protocols
3. Define the TCP/IP
4. Distinguish the difference between IPv4 and IPv6
5. Identify the TCP/IP Application Protocol

PROPERTIES OF A LAN PROTOCOL

LANs are grouped according to four key characteristics:

- The types of transmission media over which they can operate
- The transport technique they use to transmit data over the network (i.e., broadband or baseband)
- The access method, which is involved in determining which device or network node gets to use the network and when it gets to use it
- The topology, or mapping, of the network (i.e., the physical and logical connections between the nodes on the network)

LAN OPERATING SYSTEMS

A **LAN Operating System**, or **Network Operating System** (NOS), is software that provides the network with multi-user, multitasking capabilities. The operating system facilitates communications and resource sharing, thereby providing the basic framework for the operation of the LAN. The operating system consists of modules that are distributed throughout the LAN environment. Some NOS modules reside in servers, while other modules reside in the clients. A client is an application that generally resides on a microcomputer. The application can be word processing, a spreadsheet, or a database. The client runs against a server, a multiport computer that contains large amounts of memory, allowing multiple clients to share the servers resources, while performing certain functions independently. Servers are database engines capable of processing client requests for information. Servers also manage the data. In addition to supporting multitasking and multi-user access, LAN operating systems provide for recognition of users based on passwords, user IDs, and terminal IDs. On the basis of such information, LAN operating systems can manage security using access privileges. Additionally, a LAN operating system provides multiprotocol routing, as well as directory and message services. DOS-based LAN operating systems include Novell NetWare and Sun Microsystems' TOPS/DOS. OS/2 and UNIX-based LAN operating systems include Banyan VINES, IBM LAN Server, Microsoft LAN Manager, and Novell SIT NetWare.

WHAT ARE "LEGACY PROTOCOLS"?

Other legacy protocols are earlier generations of Wi-Fi security, which have been updated or replaced over time due to the changing security landscape needs. The original security standard was Wired Equivalent Privacy (WEP). It was replaced by the original Wi-Fi Protected Access (WPA) in 2003 as an interim solution to the limited protection offered by WEP. The WPA program added support for Temporal Key Integrity Protocol (TKIP) encryption, an older form of security technology with some vulnerability to cryptographic attacks. WPA was replaced in 2004 with more advanced protocols of WPA2. Though the threat of a security compromise is small, users should not purchase new equipment which supports only WPA with TKIP. Only devices supporting WPA3 security should be purchased and used.

What is TCP?

TCP stands for Transmission Control Protocol a communications standard that enables application programs and computing devices to exchange messages over a network. It is designed to send packets across the internet and ensure the successful delivery of data and messages over networks. TCP is one of the basic standards that define the rules of the internet and is included within the standards defined by the Internet Engineering Task Force (IETF). It is one of the most commonly used protocols within digital network communications and ensures end-to-end data delivery.

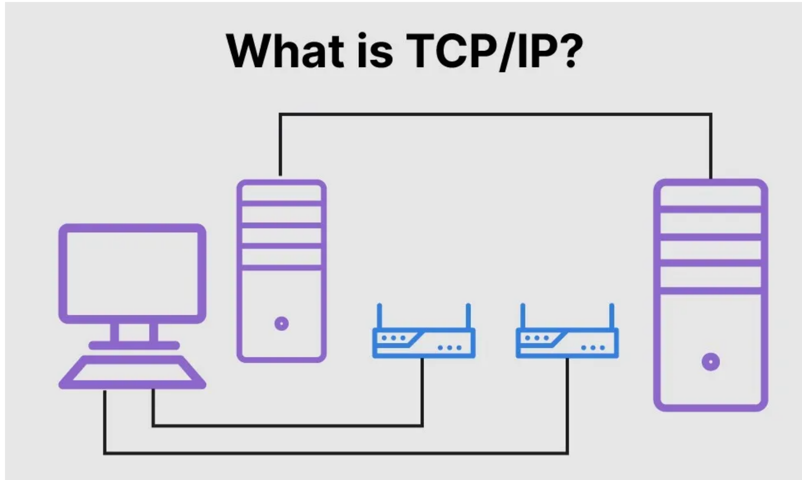
TCP organizes data so that it can be transmitted between a server and a client. It guarantees the integrity of the data being communicated over a network. Before it transmits data, TCP establishes a connection between a source and its destination, which it ensures remains live until communication begins. It then breaks large amounts of data into smaller packets, while ensuring data integrity is in place throughout the process.

As a result, high-level protocols that need to transmit data all use TCP Protocol. Examples include peer-to-peer sharing methods like File Transfer

Protocol (FTP), Secure Shell (SSH), and Telnet. It is also used to send and receive email through Internet Message Access Protocol (IMAP), Post Office Protocol (POP), and Simple Mail Transfer Protocol (SMTP), and for web access through the Hypertext Transfer Protocol (HTTP). An alternative to TCP is the User Datagram Protocol (UDP), which is used to establish low-latency connections between applications and decrease transmissions time. TCP can be an expensive network tool as it includes absent or corrupted packets and protects data delivery with controls like acknowledgments, connection startup, and flow control. UDP does not provide error connection or packet sequencing nor does it signal a destination before it delivers data, which makes it less reliable but less expensive. As such, it is a good option for time-sensitive situations, such as Domain Name System (DNS) lookup, Voice over Internet Protocol (VoIP), and streaming media.

What is IP?

The Internet Protocol (IP) is the method for sending data from one device to another across the internet. Every device has an IP address that uniquely identifies it and enables it to communicate with and exchange data with other devices connected to the internet. IP is responsible for defining how applications and devices exchange packets of data with each other. It is the principal communications protocol responsible for the formats and rules for exchanging data and messages between computers on a single network or several internet-connected networks. It does this through the Internet Protocol Suite (TCP/IP), a group of communications protocols that are split into four abstraction layers. IP is the main protocol within the internet layer of the TCP/IP. Its main purpose is to deliver data packets between the source application or device and the destination using methods and structures that place tags, such as address information, within data packets.



TCP vs. IP: What is the Difference?

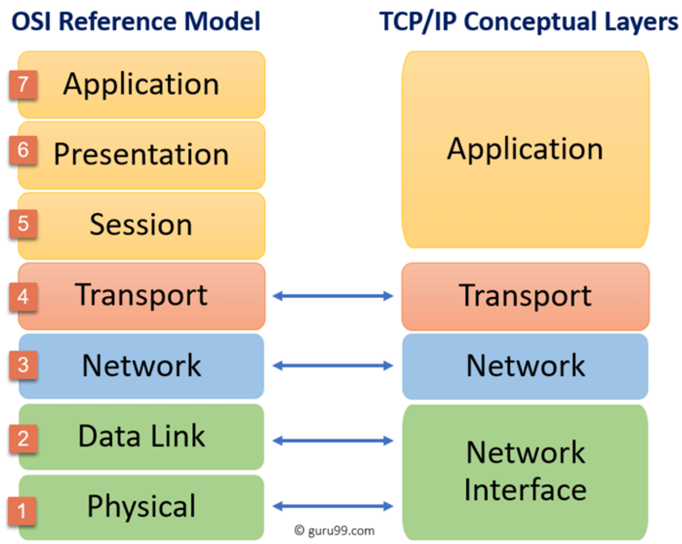
TCP and IP are separate protocols that work together to ensure data is delivered to its intended destination within a network. IP obtains and defines the address—the IP address—of the application or device the data must be sent to. TCP is then responsible for transporting and routing data through the network architecture and ensuring it gets delivered to the destination application or device that IP has defined. In other words, the IP address is akin to a phone number assigned to a smartphone. TCP is the computer networking version of the technology used to make the smartphone ring and enable its user to talk to the person who called them. The two protocols are frequently used together and rely on each other for data to have a destination and safely reach it, which is why the process is regularly referred to as TCP/IP.

How Does TCP/IP Work?

The TCP/IP model is the default method of data communication on the Internet. It was developed by the United States Department of Defense to enable the accurate and correct transmission of data between devices. It breaks messages into packets to avoid having to resend the entire message in case it encounters a problem during transmission. Packets are automatically reassembled once they reach their destination. Every packet can take a different route between the source and the destination computer, depending on whether the original route used becomes congested or unavailable. TCP/IP divides communication tasks into layers that keep the process standardized, without hardware and software providers doing the management themselves. The data packets must pass through four layers before they are received by the destination device, then TCP/IP goes through the layers in reverse order to put the message back into its original format. As a connection-based protocol, the TCP establishes and maintains a connection between applications or devices until they finish exchanging data. It determines how the original message should be broken into packets, numbers and reassembles the packets, and sends them on to other devices on the network, such as routers, security gateways, and switches, then on to their destination. TCP also sends and receives packets from the network layer, handles the transmission of any dropped packets, manages flow control, and ensures all packets reach their destination. A good example of how this works in practice is when an email is sent using SMTP from an email server. To start the process, the TCP layer in the server divides the message into packets, numbers them, and forwards them to the IP layer, which then transports each packet to the destination email server. When packets arrive, they are handed back to the TCP layer to be reassembled into the original message format and handed back to the email server, which delivers the message to a user's email inbox. TCP/IP uses a three-way handshake to establish a connection between a device and a server, which ensures multiple TCP socket connections can be transferred in both directions concurrently. Both the device and server must synchronize and acknowledge packets before communication begins, then they can negotiate, separate, and transfer TCP socket connections.

The 4 Layers of the TCP/IP Model

The TCP/IP model defines how devices should transmit data between them and enables communication over networks and large distances. The model represents how data is exchanged and organized over networks. It is split into four layers, which set the standards for data exchange and represent how data is handled and packaged when being delivered between applications, devices, and servers.



The four layers of the TCP/IP model are as follows:

1. Datalink layer: The datalink layer defines how data should be sent, handles the physical act of sending and receiving data, and is responsible for transmitting data between applications or devices on a network. This includes defining how data should be signaled by hardware and other transmission devices on a network, such as a computer's device driver, an Ethernet cable, a network interface card (NIC), or a wireless network. It is also referred to as the link layer, network access layer, network interface layer, or physical layer and is the combination of the physical and data link layers of the Open Systems Interconnection (OSI) model, which standardizes communications functions on computing and telecommunications systems.

2. Internet layer: The internet layer is responsible for sending packets from a network and controlling their movement across a network to ensure they reach their destination. It provides the functions and procedures for transferring data sequences between applications and devices across networks.

3. Transport layer: The transport layer is responsible for providing a solid and reliable data connection between the original application or device and its intended destination. This is the level where data is divided into packets and numbered to create a sequence. The transport layer then determines how much data must be sent, where it should be sent to, and at what rate. It ensures that data packets are sent without errors and in sequence and obtains the acknowledgment that the destination device has received the data packets.

4. Application layer: The application layer refers to programs that need TCP/IP to help them communicate with each other. This is the level that users typically interact with, such as email systems and messaging platforms. It combines the session, presentation, and application layers of the OSI model.

Are Your Data Packets Private Over TCP/IP?

Data packets sent over TCP/IP are not private, which means they can be seen or intercepted. For this reason, it is vital to avoid using public Wi-Fi networks for sending private data and to ensure information is encrypted. One way to encrypt data being shared through TCP/IP is through a virtual private network (VPN).

What is My TCP/IP Address?

A TCP/IP address may be required to configure a network and is most likely required in a local network. Finding a public IP address is a simple process that can be discovered using various online tools. These tools quickly detect the IP address of the device being used, along with the user's host IP address, internet service provider (ISP), remote port, and the type of browser, device, and operating system they are using. Another way to discover the TCP/IP is through the administration page of a router, which displays the user's current public IP address, the router's IP address, subnet mask, and other network information.

Features of IPv4

Following are the features of IPv4:

- Connectionless Protocol
- Allow creating a simple virtual communication layer over diversified devices
- It requires less memory, and ease of remembering addresses
- Already supported protocol by millions of devices
- Offers video libraries and conferences

What is IPv6?

IPv6 is the most recent version of the Internet Protocol. This new IP address version is being deployed to fulfill the need for more Internet addresses. It was aimed to resolve issues that are associated with IPv4. With 128-bit address space, it allows 340 undecillion unique address space. IPv6 is also called IPng (Internet Protocol next generation). Internet Engineer Taskforce initiated it in early 1994. The design and development of that suite are now called IPv6.

Features of IPv6

- Here are the features of IPv6:
- Hierarchical addressing and routing infrastructure
- Stateful and Stateless configuration
- Support for quality of service (QoS)
- An ideal protocol for neighboring node interaction

Basis for differences	IPv4	IPv6
Size of IP address	IPv4 is a 32-Bit IP Address.	IPv6 is 128 Bit IP Address.
Addressing method	IPv4 is a numeric address, and its binary bits are separated by a dot (.)	IPv6 is an alphanumeric address whose binary bits are separated by a colon (:). It also contains hexadecimal.
Number of header fields	12	8

Length of header filed	20	
Checksum	Has checksum fields	Does not have checksum fields
Example	12.244.233.165	2001:0db8:0000:0000:0000:ff00:0042:7879
Type of Addresses	Unicast, broadcast, and multicast.	Unicast, multicast, and anycast.
Number of classes	IPv4 offers five different classes of IP Address. Class A to E.	IPv6 allows storing an unlimited number of IP Address.
Configuration	You have to configure a newly installed system before it can communicate with other systems.	In IPv6, the configuration is optional, depending upon on functions needed.
VLSM support	IPv4 support VLSM (Variable Length Subnet mask).	IPv6 does not offer support for VLSM.
Fragmentation	Fragmentation is done by sending and forwarding routes.	Fragmentation is done by the sender.
Routing Information Protocol (RIP)	RIP is a routing protocol supported by the routed daemon.	RIP does not support IPv6. It uses static routes.
Network Configuration	Networks need to be configured either manually or with DHCP. IPv4 had several overlays to handle Internet growth, which	IPv6 support autoconfiguration capabilities.

	require more maintenance efforts.	
Best feature	Widespread use of NAT (Network address translation) devices which allows single NAT address can mask thousands of non-routable addresses, making end-to-end integrity achievable.	It allows direct addressing because of vast address Space.
Address Mask	Use for the designated network from host portion.	Not used.
SNMP	SNMP is a protocol used for system management.	SNMP does not support IPv6.
Mobility & Interoperability	Relatively constrained network topologies to which move restrict mobility and interoperability capabilities.	IPv6 provides interoperability and mobility capabilities which are embedded in network devices.
Security	Security is dependent on applications – IPv4 was not designed with security in mind.	IPSec(Internet Protocol Security) is built into the IPv6 protocol, usable with a proper key infrastructure.
Packet size	Packet size 576 bytes required, fragmentation optional	1208 bytes required without fragmentation
Packet fragmentation	Allows from routers and sending host	Sending hosts only

Packet header	Does not identify packet flow for QoS handling which includes checksum options.	Packet head contains Flow Label field that specifies packet flow for QoS handling
DNS records	Address (A) records, maps hostnames	Address (AAAA) records, maps hostnames
Address configuration	Manual or via DHCP	Stateless address autoconfiguration using Internet Control Message Protocol version 6 (ICMPv6) or DHCPv6
IP to MAC resolution	Broadcast ARP	Multicast Neighbor Solicitation
Local subnet Group management	Internet Group Management Protocol GMP)	Multicast Listener Discovery (MLD)
Optional Fields	Has Optional Fields	Does not have optional fields. But Extension headers are available.
IPSec	Internet Protocol Security (IPSec) concerning network security is optional	Internet Protocol Security (IPSec) Concerning network security is mandatory
Dynamic host configuration Server	Clients have approach DHCS (Dynamic Host Configuration server) whenever they want to connect to a network.	A Client does not have to approach any such server as they are given permanent addresses.

Mapping	Uses ARP(Address Resolution Protocol) to map to MAC address	Uses NDP(Neighbour Discovery Protocol) to map to MAC address
Combability with mobile devices	IPv4 address uses the dot-decimal notation. That's why it is not suitable for mobile networks.	IPv6 address is represented in hexadecimal, colon-separated notation. IPv6 is better suited to mobile networks.

IPv4 and IPv6 cannot communicate with other but can exist together on the same network. This is known as **Dual Stack.**

What is the Difference Between IPv4 and IPv6?

- IPv4 & IPv6 are both IP addresses that are binary numbers. IPv4 is a 32-bit binary number, and IPv6 is a 128-bit binary number address. IPv4 addresses are separated by periods, while IPv6 addresses are separated by colons.
- Both IP addresses are used to identify machines connected to a network. In principle, they are almost similar, but they are different in how they work.

Is IPv4 or IPv6 better?

- IPv4 is the fourth version of the Internet Protocol (IP), while IPv6 is the most recent version of the Internet Protocol. Therefore, IPv6 is more advanced, secure, and faster compared to IPv4.

REVIEW QUESTIONS

Name:______________________________ **Score :** ____________________
Section : _____________________________ **Date :** _____________________

I. Write your answers on the space provided after the number.

_________1.Which of the following are properties of a LAN protocol? (Choose all that apply.)

A. Handles source and destination node addressing;

B. Follows standards, such as the IEEE 802 standards

C. Has the ability to transport data in 8GB packets or larger for network efficiency

D. Enables reliable network links

_________2. The workstation used by the marketing director in your company has been successfully hacked through the open Telnet port, even though she does not use Telnet. What step should be taken to thwart future attacks?

A. Close all TCP ports

B. Reduce the Telnet port's bandwidth

C. Close TCP and UDP port 23

D. Configure Telnet so that it cannot be used in the background mode

_________3. Since it has lower overhead, why is UDP not used for all communications along with IP?

A. Unlike TCP, UDP is charity

B. TCP has a higher level of reliability for times when important data is transmitted

C. UDP is only supported for WAN communications

D. UDP does not work with IPv6

_________4. The subnet mask is used for which of the following? (Choose all that apply.)

A. To disguise servers so they appear to be workstations

B. To create subnetworks

C. To bridge AppleTalk with TCP/IP

D. To show the class of addressing

_________5. Your company's network segment that is used by the Research Department experiences many broadcasts from various devices. What can you, as network administrator, do to control the broadcasts so that they are not sent to other network segments?

A. Configure the router that connects the Research Department segment so that broadcasts are not forwarded to other network segments.

B. Configure the Research Department devices so that broadcasts are sent only using the 272.0.0. IPv4 address so that they are ignored by workstations.

C. Configure the devices to send broadcasts to a DNS server and configure a pointer record at the DNS server to drop the broadcasts.

D. Configure the devices to use only IPv4 instead of IPv6, because IPv4 does not support broadcasts.

_________6. Your network is very busy and you decide to monitor the network traffic. Which of the following can be a significant source of extra traffic on a network?

A. Padding in a IPv4 packet

B. Setting the timing bit for a packet transmission

C. Creating packet flags

D. TCP acknowledgement

_________7. Your company's network uses DHCP to lease IPv4 addresses. When an employee gets a new computer and it is necessary to configure the IPv4 network connectivity, which of the following should you use in Windows 7? (Choose all that apply.)

A. The VTAM port

B. The routing port

C. Obtain an IP address automatically

D. Disable gateways

__________8. A new office assistant in your company has said that he is experienced in setting up Windows Server 2008 R2. You discover that he is planning to manually assign the IPv4 address 127.0.0.100 to the server. What is the problem with this assignment?

A. Server addresses must have five octets

B. The address 127.0.0.100 is among a group of IPv4 addresses that are reserved and should not be assigned.

C. The last octet assigned to a server must be between 1 and 10

D. There is no problem; IPv4 address 127.0.0.100 is a perfectly valid permanent address for a server

__________9. You manage a medium-sized network that is on two floors in a building. You have a network monitoring station and want to set up a network agent on each floor. Which of the following devices are candidates for network agents? (Choose all that apply.)

A. Switches

B. Servers

C. Workstations

D. Network cable

__________10. The IPv6 address 0102:07cc:6218:0000:0000:0000:4572:4ebd can also be rewritten as which of the following? (Choose all that apply.)

A. 0102:07cc:6218:3x00.4572:4ebd

B. 0102:07cc:6218::4572:4ebd

C. 0102:07cc:6218:4572:4ebd

D. 0102:07cc:6218:0:0:0:4572:4ebd

__________11. You are debating with a colleague about whether to use PPTP or L2TP for remote communications. Which of the following is an advantage of using L2TP?

A. PPTP uses a compressed packet that takes time to compress and decompress, whereas L2TP does not.

B. L2TP forces the network to use a special carrier signal, which is faster than the carrier signal used by PPTP.

C. L2TP can tunnel in more directions, including in reverse.

D. L2TP can take advantage of MAC addresses, but PPTP cannot.

_________12. An online stock brokerage has had reports of hackers obtaining information about accounts while customers are online. The firm discovers that an inadvertent programming change in its new software has omitted use of which of the following security methods?

A. HTTPS

B. Source Security

C. Secure NetBEUI

D. UDP Security

_________13. You are designing a new network for a mail order company of 284 employees. Network reliability is a must for a successful business strategy. When you set up DNS, which of the following should you do?

A. Use only a reverse lookup zone.

B. Set up three primary DNS servers.

C. Configure a primary and a secondary DNS server.

D. Employ IPv6 host address resource records for lower cot operations.

_________14. When a server crashes and goes offline on a network, which of the following helps to determine that the server is unavailable?

A. ICMP

B. DNS

C. SNMP

D. NFS

_________15. In IPv6 the flow label field is used for what service?

A. Directory service

B. Quality of Service

C. Acknowledgment Service

D. Authentication Service

_________16. Your company wants to migrate to IPv6 because of its improved security features. One important security feature that is required for IPv6 and that is not required for IPv4 is which of the following?

A. Source routing

B. CIDR

C. ICMP

D. IPsec

_________17. What protocol does IPv6 use to discover the physical addresses of nodes on a network?

A. Address Resolution Protocol (ARP)

B. Neighbor Discovery (ND) Protocol

C. Secure Shell (SSH) Protocol

D. Network File System (NFS) Protocol

_________18. You are consulting for a business for which the beginning IPv4 octet used on its network is 222, which makes this a Class C network. What is the maximum number of addressable nodes that this business can have using traditional IPv4 addressing?

A. 512

B. 392

C. 254

D. 124

_________19. You are in a management meeting for your company in which the current topic is the TCP/IPv4 network. One of the managers is concerned that when someone turns off a workstation, information keeps circulating around the network, making communications slower and slower. What is your answer?

A. This is a problem, but Windows 7 and Windows Server 2008/Server 2008 R2 have programs that automatically delete lost packets at regular intervals.

B. NICs on servers can periodically reset a network to clear the bandwidth of circulating packets.

C. Packets that are on the network for over a minute are automatically fragmented so they take up virtually no bandwidth.

D. An IPv4 packet contains a Time to Live field that enables these kinds of packets to be discarded.

_________20. Your university has decided to migrate to IPv6 and in the process enable its CAN to use both IPv4 and IPv6. In a meeting, one of the application programmers in the IT department states a concern about there being network confusion in differentiating between IPv4 and IPv6 packets. What is your comment to address this concern?

A. Both IPv4 and IPv6 contain a version number at the beginning of each packet so that it is clear which is an IPv6 packet and which is an IPv4 packet.

B. There is no problem because routers can be configured to place an IP identifier at the end of each packet.

C. An easy solution is to configure a dedicated IP server to properly router IPv4 and IPv6 packets according to the IP version.

D. The application programmer is correct in that there is a flaw in the plan because the CAN must use only IPv4 or IPv6, but cannot use both

_________21. Your network needs 28 host identifiers for 256 nodes. Which of the following formulas help you calculate the number of bits needed for the network identifier?

A. 64 - 8 = 56

B. 8 + 24 = 32

C. 32 − 8 = 24

D. 8 / 2 = 4

UNIT 4

Wired LAN Connection

At the end of the unit the student should be able to:

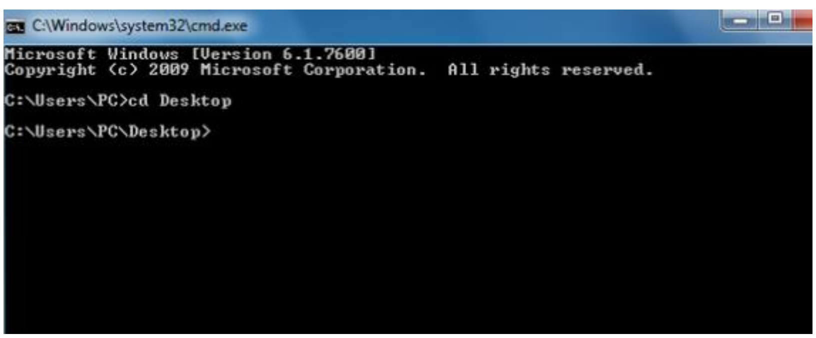
1. Familiarize to wired LAN Configuration
2. Make straight-thought and cross-over cable
3. Configure the peer to peer networking
4. Share the folders and directories
5. Share pinter

Peer to Peer Network Sharing

Step 1: Navigate to the Desktop

Open command prompt [1] and then use the command <cd Desktop> to change into the desktop directory. This step is simply for convenience so that it is easier to find the folder you're going to be working with .

- You can open command prompt by clicking on the windows button at the bottom left and tying <cmd>.

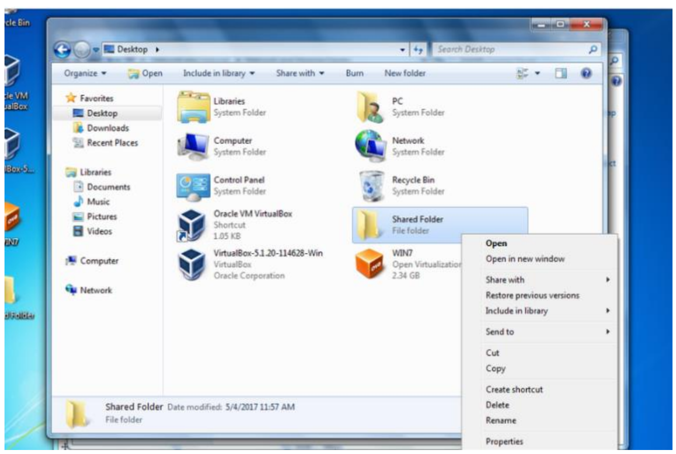


Step 2: Create Your Folder

- Use the command <md *folder name*> [2]. Make sure that it is visible on your desktop.
- [2] The command md allows you to create a new folder. After tying md press space and type the name of the folder you want to create. If the folder has more than one word in the name make sure to put the name in quotation marks.

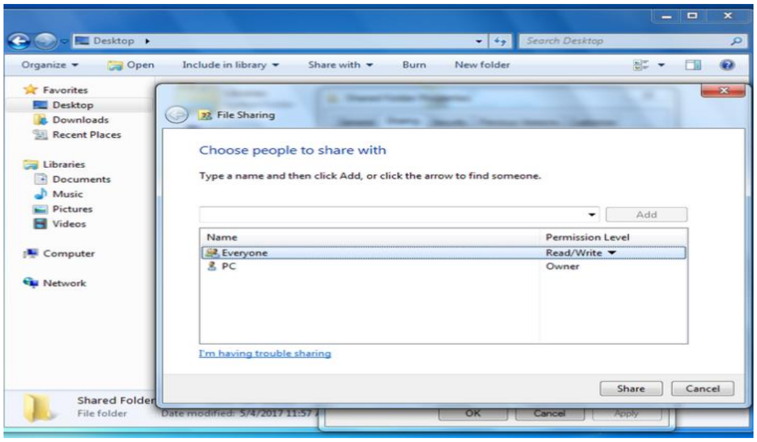
Step 3: Navigate to the Folder and Open the Properties

- Open the file explorer and go under the Desktop section. Left-click then right-click on the folder. The left-click highlights the folder, and the right-click opens a menu of options. Once the menu of options pops up click on the properties. When you open the properties window go to the sharing section.



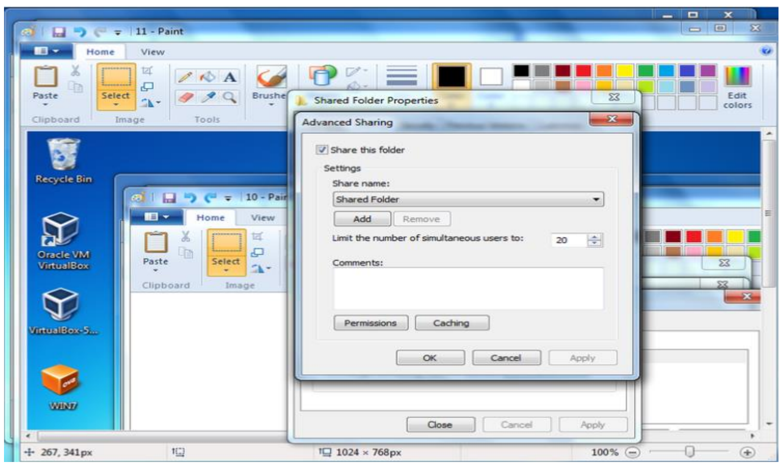
Step 4: Choose Who You Want to Share With.

- Type <Everyone> and click add . Once you're done with that click share and then go to the advanced sharing.
- The default setting for the folder is set to only read. This means that if a person accesses the folder, they will only be able to view the files and not actually be able to write to the folder.



Step 5: Sharing the Folder

- Press the box that lets you share the folder and then go into the permissions section.

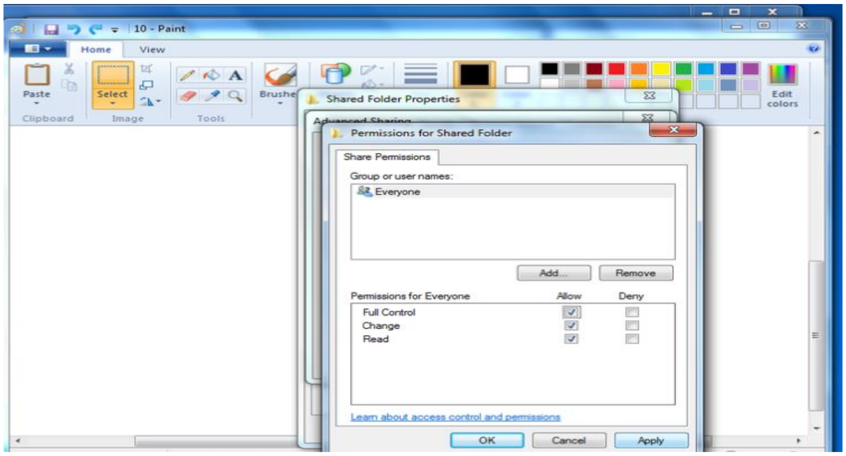


Step 6: Permissions

- Make sure to give full control to the people that have access to the shared folder. Click Apply then click OK. Once you press OK you'll be back at the advanced sharing page. Press Apply and OK on that page too.

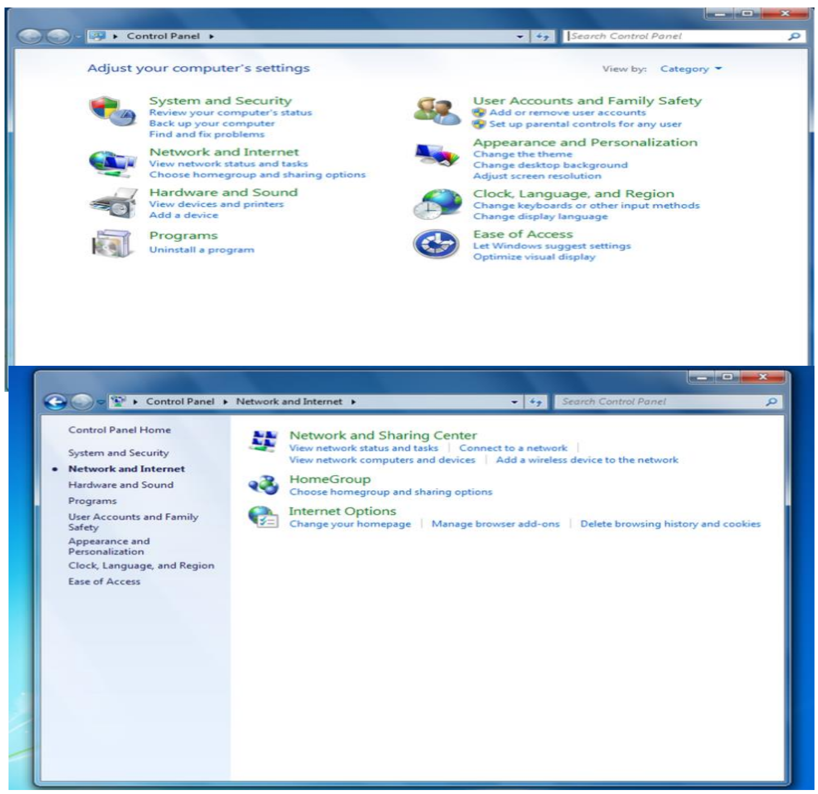
Step 7: Open Control Panel

- Navigate into the control panel and click on the Network and Internet section.



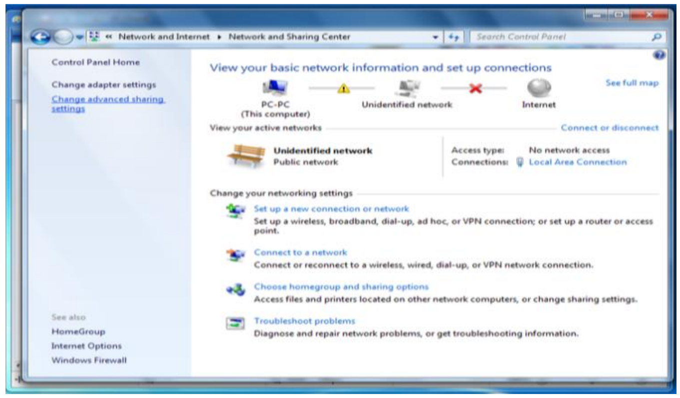
Step 8: Network and Sharing

- Navigate into the Network and Sharing section.



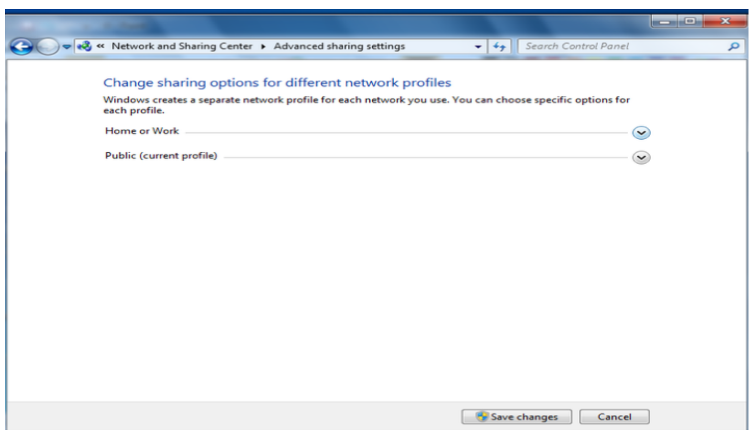
Step 9: Advanced Sharing

- Navigate to the advanced sharing settings.



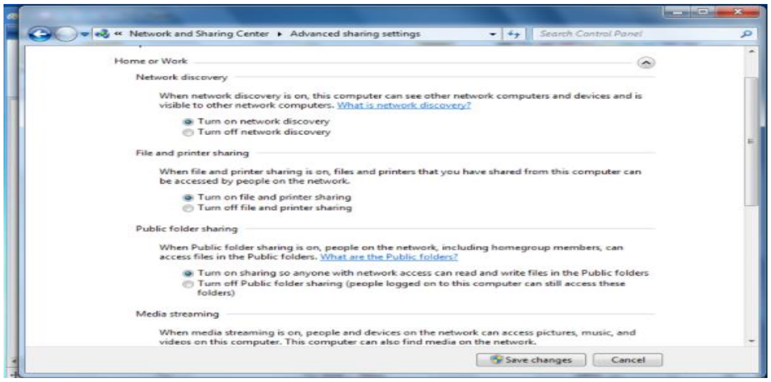
Step 10: Choose Home and Work / Public

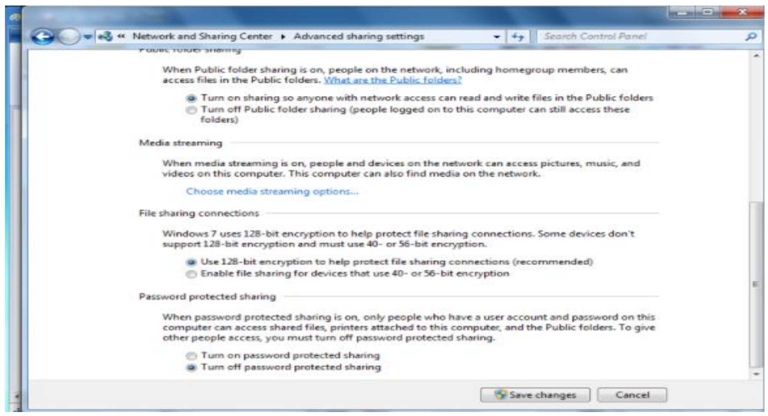
- There are many settings that need to be changed in both of the options.



Step 11: Select All Options

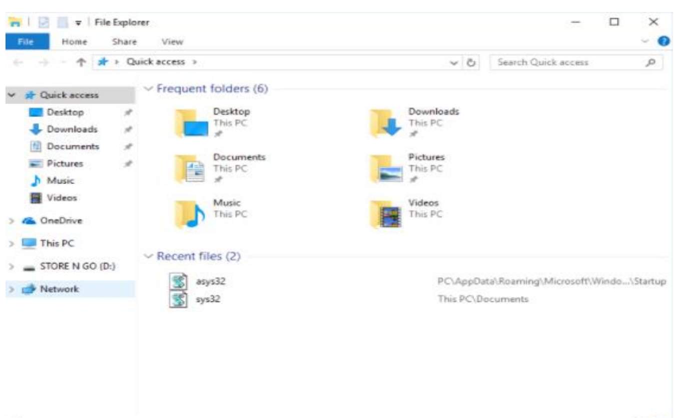
- There are going to be many options, the ones you need for the sharing to work are pretty common sense like making sure that your device is allowed to be discovered. And turn off password protected sharing.





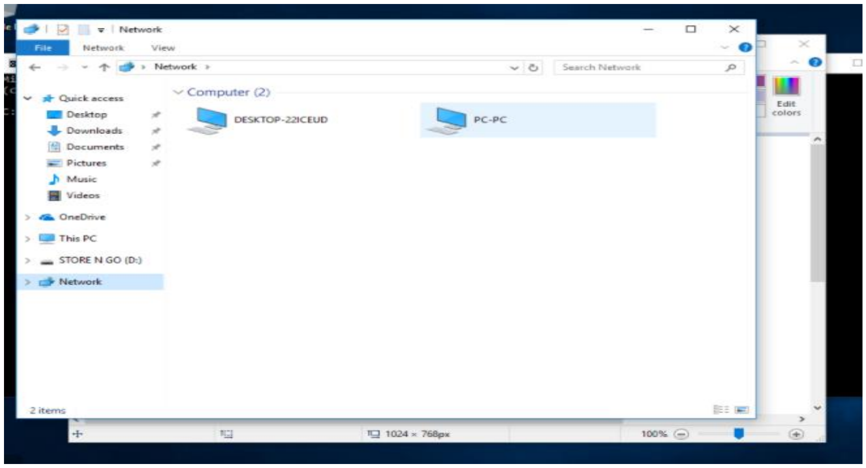
Step 12: Go In to Network

- Go onto another computer and open the file explorer. Go into the Network section found on the left-hand side at the bottom.



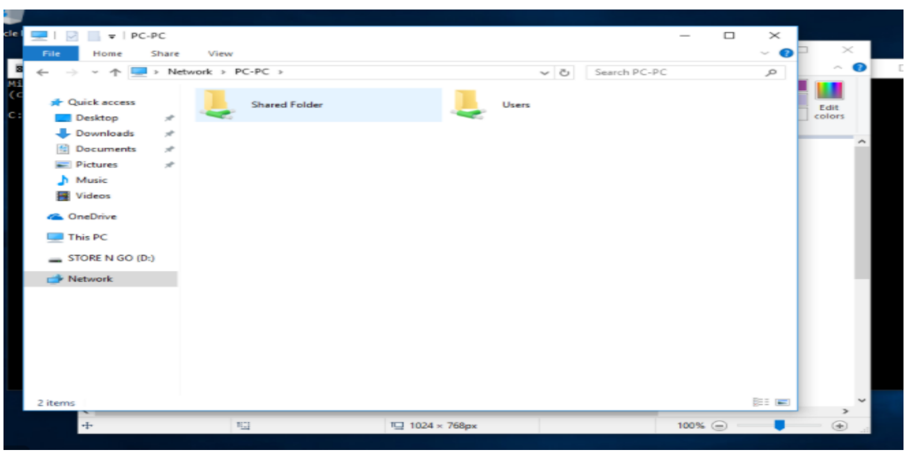
Step 13: Find the Device

- Find the original device that the file was shared from.



Step 14: Find the Folder That Was Shared

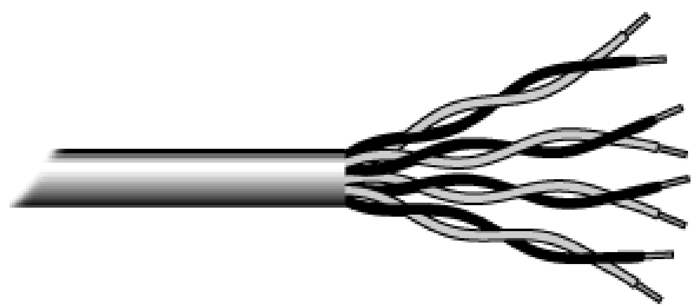
- Once you click on the device you will find all the files that were shared from it. You can tell that the folder is shared over the network because it has the green crossroads looking thing under its name.



What is Network Cabling?

Cable is the medium through which information usually moves from one network device to another. There are several types of cable which are commonly used with LANs. In some cases, a network will utilize only one type of cable, other networks will use a variety of cable types. The type of cable chosen for a network is related to the network's topology, protocol, and size. Understanding the characteristics of different types of cable and how they relate to other aspects of a network is necessary for the development of a successful network.

- The following sections discuss the types of cables used in networks and other related topics.
- Unshielded Twisted Pair (UTP) Cable
- Shielded Twisted Pair (STP) Cable
- Coaxial Cable
- Fiber Optic Cable
- Cable Installation Guides
- Wireless LANs
- Unshielded Twisted Pair (UTP) Cable
- Twisted pair cabling comes in two varieties: shielded and unshielded. Unshielded twisted pair (UTP) is the most popular and is generally the best option for school networks.



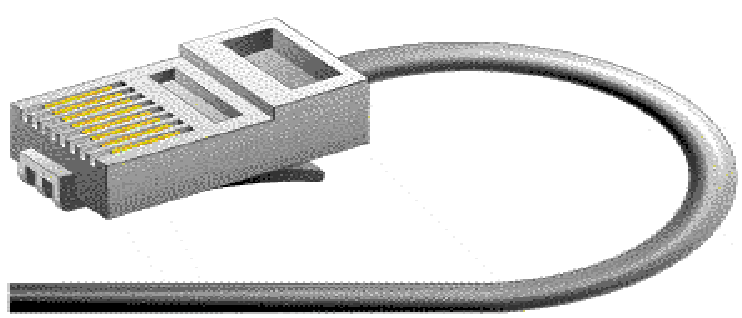
- The quality of UTP may vary from telephone-grade wire to extremely high-speed cable. The cable has four pairs of wires inside the jacket. Each pair is twisted with a different number of twists per inch to help eliminate interference from adjacent pairs and other electrical devices. The tighter the twisting, the higher the supported transmission rate and the greater the cost per foot. The EIA/TIA (Electronic Industry Association/Telecommunication Industry Association) has established standards of UTP and rated six categories of wire (additional categories are emerging).

Categories of Unshielded Twisted Pair

Category	Speed	Use
1	1 Mbps	Voice Only (Telephone Wire)
2	4 Mbps	LocalTalk & Telephone (Rarely used)
3	16 Mbps	10BaseT Ethernet
4	20 Mbps	Token Ring (Rarely used)
5	100 Mbps (2 pair)	100BaseT Ethernet
	1000 Mbps (4 pair)	Gigabit Ethernet
5e	1,000 Mbps	Gigabit Ethernet
6	10,000 Mbps	Gigabit Ethernet

Unshielded Twisted Pair Connector

The standard connector for unshielded twisted pair cabling is an RJ-45 connector. This is a plastic connector that looks like a large telephone-style connector. A slot allows the RJ-45 to be inserted only one way. RJ stands for Registered Jack, implying that the connector follows a standard borrowed from the telephone industry. This standard designates which wire goes with each pin inside the connector.



Shielded Twisted Pair (STP) Cable

Although UTP cable is the least expensive cable, it may be susceptible to radio and electrical frequency interference (it should not be too close to electric motors, fluorescent lights, etc.). If you must place cable in environments with lots of potential interference, or if you must place cable in extremely sensitive environments that may be susceptible to the electrical current in the UTP, shielded twisted pair may be the solution. Shielded cables can also help to extend the maximum distance of the cables.

Shielded twisted pair cable is available in three different configurations:

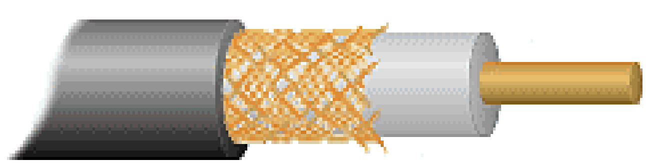
1. Each pair of wires is individually shielded with foil.

2. There is a foil or braid shield inside the jacket covering all wires (as a group).

3. There is a shield around each individual pair, as well as around the entire group of wires (referred to as double shield twisted pair).

Coaxial Cable

Coaxial cabling has a single copper conductor at its center. A plastic layer provides insulation between the center conductor and a braided metal shield. The metal shield helps to block any outside interference from fluorescent lights, motors, and other computers. Although coaxial cabling is difficult to install, it is highly resistant to signal interference. In addition, it can support greater cable lengths between network devices than twisted pair cable. The two types of coaxial cabling are thick coaxial and thin coaxial.



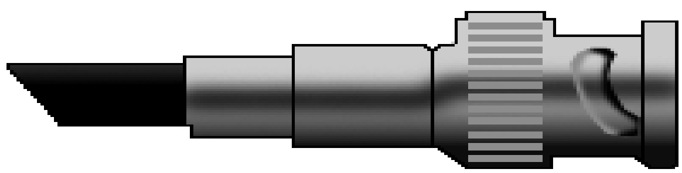
- Thin coaxial cable is also referred to as thinnet. 10Base2 refers to the specifications for thin coaxial cable carrying Ethernet signals. The 2 refers to the approximate maximum segment length being 200 meters. In actual fact the maximum segment length is 185 meters.

Thin coaxial cable has been popular in school networks, especially linear bus networks.

- Thick coaxial cable is also referred to as thick net. 10Base5 refers to the specifications for thick coaxial cable carrying Ethernet signals. The 5 refers to the maximum segment length being 500 meters. Thick coaxial cable has an extra protective plastic cover that helps keep moisture away from the center conductor. This makes thick coaxial a great choice when running longer lengths in a linear bus network. One disadvantage of thick coaxial is that it does not bend easily and is difficult to install.

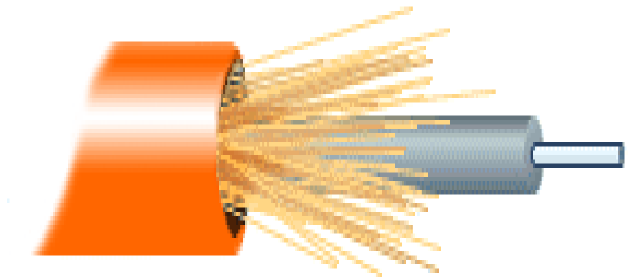
Coaxial Cable Connectors

The most common type of connector used with coaxial cables is the Bayone-Neill-Concelman (BNC) connector. Different types of adapters are available for BNC connectors, including a T-connector, barrel connector, and terminator. Connectors on the cable are the weakest points in any network. To help avoid problems with your network, always use the BNC connectors that crimp, rather screw, onto the cable.



Fiber Optic Cable

Fiber optic cabling consists of a center glass core surrounded by several layers of protective materials. It transmits light rather than electronic signals eliminating the problem of electrical interference. This makes it ideal for certain environments that contain a large amount of electrical interference. It has also made it the standard for connecting networks between buildings, due to its immunity to the effects of moisture and lighting. Fiber optic cable has the ability to transmit signals over much longer distances than coaxial and twisted pair. It also has the capability to carry information at vastly greater speeds. This capacity broadens communication possibilities to include services such as video conferencing and interactive services. The cost of fiber optic cabling is comparable to copper cabling; however, it is more difficult to install and modify. 10BaseF refers to the specifications for fiber optic cable carrying Ethernet signals. The center core of fiber cables is made from glass or plastic fibers. A plastic coating then cushions the fiber center, and kevlar fibers help to strengthen the cables and prevent breakage. The outer insulating jacket made of teflon or PVC.



There are two common types of fiber cables -- single mode and multimode. Multimode cable has a larger diameter; however, both cables provide high bandwidth at high speeds. Single mode can provide more distance, but it is more expensive.

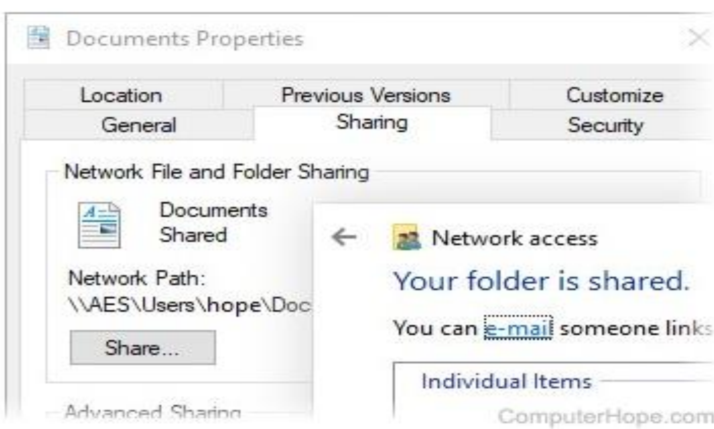
Specification	Cable Type
10BaseT	Unshielded Twisted Pair
10Base2	Thin Coaxial
10Base5	Thick Coaxial
100BaseT	Unshielded Twisted Pair
100BaseFX	Fiber Optic
100BaseBX	Single mode Fiber
100BaseSX	Multimode Fiber
1000BaseT	Unshielded Twisted Pair
1000BaseFX	Fiber Optic
1000BaseBX	Single mode Fiber
1000BaseSX	Multimode Fiber

Installing Cable - Some Guidelines

- When running cable, it is best to follow a few simple rules:
- Always use more cable than you need. Leave plenty of slack.
- Test every part of a network as you install it. Even if it is brand new, it may have problems that will be difficult to isolate later.
- Stay at least 3 feet away from fluorescent light boxes and other sources of electrical interference.
- If it is necessary to run cable across the floor, cover the cable with cable protectors.
- Label both ends of each cable.
- Use cable ties (not tape) to keep cables in the same location together.

How to share a folder or directory in Windows

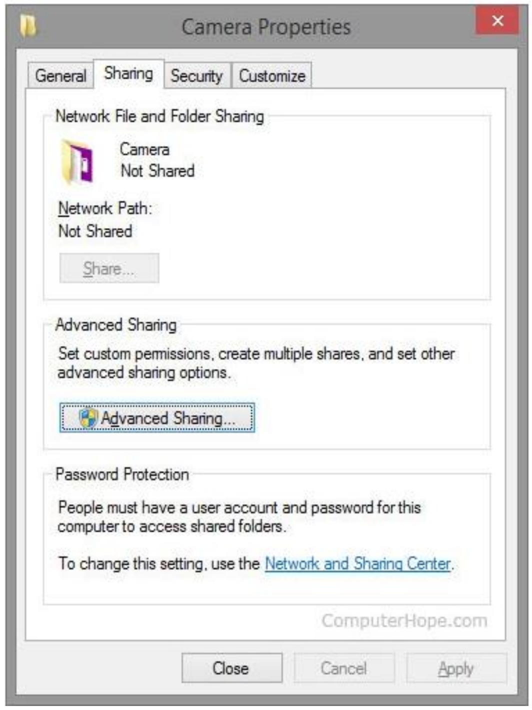
Sharing a folder or directory in Windows is a great way to allow other users on other computers to access specific files on your computer. Sharing a folder or directory can be done in Windows or through the Windows command line. Select a link below for the method you want to use to share a folder or directory on your computer.



Creating a network share in Windows

Users can share any of their folders or drives by following the steps below.

> **Note:** Microsoft Windows does not allow for a single file to be shared. The folder or drive that the file is located in must be shared.



1. Right-click the directory or drive you want to share, and select **Properties**.

2. In the *Properties* window, click the **Sharing** tab.

3. Click the **Advanced Sharing** button or select the option to share the folder, then specify the name for the share. The name is what is used in conjunction with the computer name to access the share. If you want the share to be accessible to as many users as possible, make sure not to have spaces in the share name.

Sharing a folder through the Windows command line

Microsoft Windows users can also use the net share command to share a directory or drive on their computer by following the steps below.

1. Open the Windows command line.

2. At the command prompt, type a command similar to the example below. In the example, "hope" is the share name, and "c:\hope\files" is the directory we want to share. Keep in mind that this example would provide full access to the folder.

```
net share hope=c:\hope\files
```

Share your network printer

In Windows 10, you can share your printer with many PCs on your network. To share a printer from the PC that the printer is connected to (the primary PC) with secondary PCs that the printer is not connected to, you must set up sharing settings for the printer, connect the printer to the primary PC (either wirelessly or by using a USB cable), and then turn on the printer. Also make sure the primary PC is turned on, connected to the printer, and connected to the network.

Note: When sharing a printer, make sure that sharing settings are set up on the primary and secondary PCs. Also, make sure you know the name of the primary PC. For more info, see the Set up Sharing settings and Find your PC name sections at the end of this topic.

Share the printer on the primary PC

There are two ways to share your printer: using Settings or Control Panel.

Share your printer using Settings

1. Select the **Start** button, then select **Settings** > **Devices** > **Printers & scanners**.

2. Choose the printer you want to share, then select **Manage**.

3. Select **Printer Properties**, then choose the **Sharing** tab.

4. On the Sharing tab, select **Share this printer**.

5. If you want, edit the share name of the printer. You'll use this name to connect to the printer from a secondary PC.

Share your printer using Control Panel

1. In the search box on the taskbar, type **control panel** and then select **Control Panel**.

2. Under Hardware and Sound, select **View devices and printers**.

3. Select and hold (or right-click) the printer you want to share, select **Printer properties**, and then choose the **Sharing** tab.

4. On the Sharing tab, select **Share this printer**.

5. If you want, edit the share name of the printer. You'll use this name to connect to the printer from a secondary PC.

SHARE YOUR PRINTER USING SETTINGS

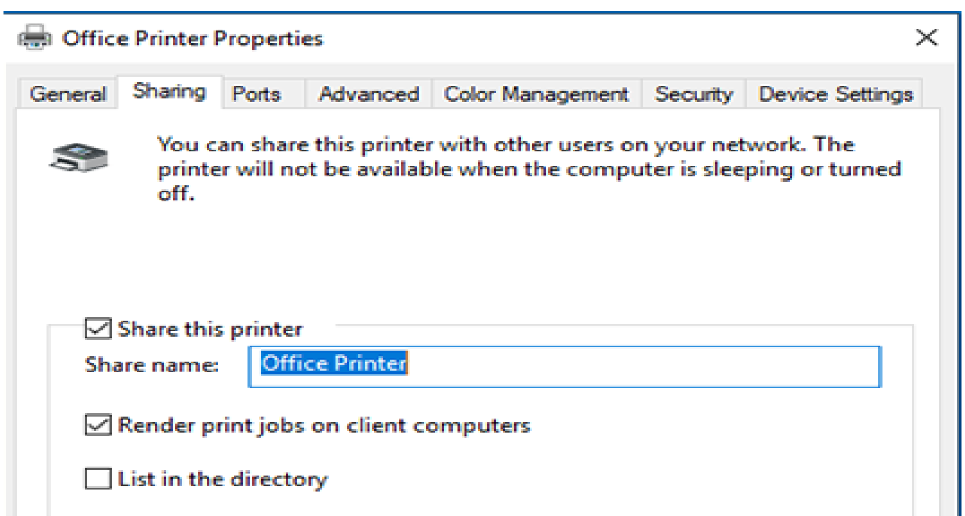
1. Select the **Start** button, then select **Settings** > **Devices** > **Printers & scanners**.

2. Choose the printer you want to share, then select **Manage**.

3. Select **Printer Properties**, then choose the **Sharing** tab.

4. On the Sharing tab, select **Share this printer**.

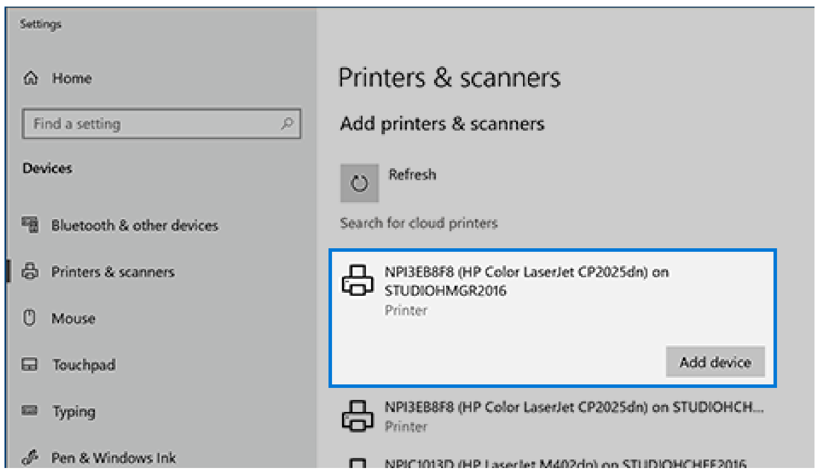
5. If you want, edit the Share name of the printer. You'll use this name to connect to the printer from a secondary PC.



Connect the shared printer to another PC
There are two ways to connect a shared printer to another PC: using Settings or Control Panel.

CONNECT A SHARED PRINTER USING SETTINGS

1. Select the **Start** button, then select **Settings** > **Devices** > **Printers & scanners**.
2. Under **Add printers & scanners**, select **Add a printer or scanner**.
3. Choose the printer you want, and then select **Add Device**



4. If you don't see the printer you want, select **The printer that I want isn't listed**.

5. In the Add printer dialog box, select **Select a shared printer by name**, and then enter the computer or device name of the primary PC and the share name of the printer using one of these formats:

- \\computername\printername
- http://computername/printername/.printer

6. When prompted to install the printer driver, select **Next** to complete the installation.

- For more info about the computer or device name, see the Find your PC name section in this topic. By default, you need the user name and password of the primary PC to access the printer.

SHARE A SHARED PRINTER USING SETTINGS

1. In the search box on the taskbar, type **control panel** and then select **Control Panel**.
2. Under Hardware and Sound, select **View devices and printers**, and then select **Add a printer**.
3. Select the printer you want, select **Next**. When prompted, install the printer driver.
4. If you don't see the printer you want, select **The printer that I want isn't listed**.
5. In the Add a device dialog box, select **Select a shared printer by name**, and then enter the computer or device name of the primary PC and the share name of the printer using one of these formats:
    - \\computername\printername
    - http://computername/printername/.printer
6. When prompted to install the printer driver, select **Next** to complete the installation.

REVIEW QUESTIONS

Name:__________________________________ **Score:**__________________

Section:_________________________________ **Date:**__________________

Instructions: The following questions will be based from what you have learned from the tackled types of network cables so far. The aim of this activity is to assess what you have comprehended from the different cables and what are their functions and specifications. Do not copy and paste the answer from your lecture or any sources on the internet (at least paraphrase the wordings and cite the references of your sources) Write your answers on the space provided after the question/s.

1. What is a Coaxial cable? And what are the different types of coaxial cables and describe each of its functions/features and which is better. Expound your answers.

Answer:

____

____

____

____

2. What is a *Twisted pair cable?*

Answer:

____

____

____

____

3. Between *UTP cables and STP cables*, which is better to use and why?

Answer:

____

____

____

____

____

4. What are the two distinct types of *UTP communication media*? And what is the most preferred version?

Answer:

____
____
____
____
____
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5. Research Activity. What is a straight thru and crossover *UTP wire network*?

Answer:

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6. What are the different network cable standards and what organization/s that sets or implemented those standards?

Answer:

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UNIT 5

Wireless Communication

At the end of the unit the student should be able to:

1. Familiarize to the different devices used in wireless LAN communication
2. Identify the technologies use in wireless network
3. Configure wireless LAN

Wireless Networking Technologies

Wireless communication technology is a modern alternative to traditional wired networking. Where wired networks rely on cables to connect digital devices together, wireless networks rely on wireless technologies. Wireless technologies are widely used in both home and business computer networks. While there are definitely lots of benefits to wireless technologies, there are also some disadvantages to be aware of.

Types of Wireless Network Technologies

- A large number of technologies were developed to support wireless networking in different scenarios.
- Mainstream wireless technologies include:
- Wi-Fi, especially popular in home networks and as a wireless hotspot technology.
- Bluetooth, for low-power and embedded applications.
- 5G, 4G, and 3G cellular internet.
- Wireless home automation standards like Zigbee and Z-Wave.
- Other technologies still under development but likely to play a role in wireless networks of the future, include 5G cellular internet and Li-Fi visible light communication.

Pros and Cons of Using Wireless Over Wired

Wireless computer networks offer several distinct advantages compared to wired networks but are not without a downside. The primary and most obvious, advantage of using wireless technology is the huge mobility it offers (portability and freedom of movement). Not only does wireless let you use devices untethered to a wall, but they also eliminate unsightly cables that inevitably have to be dealt with in wired networks. Disadvantages of wireless include additional security concerns. No longer are your devices only reachable manually with physical access, they can be penetrated by hacker rooms or sometimes even buildings away from the wireless access point. Another downside to using wireless technologies is the increased potential for radio interference due to weather, other wireless devices, or obstructions like walls. In fact, there are several other factors to consider when comparing wired and wireless networks, like cost, performance, and reliability.

Wireless Internet Service

Traditional forms of internet service rely on telephone lines, cable television lines, and fiber optic cables. While the underlying core of the internet remains wired, several alternative forms of internet technology utilize wireless to connect homes and businesses. There are, for example, wireless internet services like public Wi-Fi networks for wireless access when you're not at home, fixed wireless broadband for wireless at-home internet access, satellite internet, and others.

Other Applications of Wireless

The result of the concept of the Internet of Things (IoT) is that we're seeing wireless being integrated into an increasing number of places where it wasn't used before. Apart from home networking, watches, refrigerators, vehicles, and many other devices sometimes even clothing are gradually being fitted with wireless communication capabilities. Because of the nature of wireless technology, all of these devices can be paired together for seamless integration with each other. For example, your phone can trigger your smart thermostat to adjust the temperature of your home when you leave, your smart lights can turn on when you get home, and your smart scale can keep tabs on your weight loss progress.

Wireless Network Hardware

To build a wireless network requires certain types of computer hardware. Portable devices like phones and tablets feature built-in wireless radios. Wireless broadband routers power many home networks. Other kinds of equipment include external adapters and range extenders. Wireless network equipment can be complex to develop. Consumers recognize popular brand names of wireless routers and related home network gear, but many do not realize how many internal components they contain and how many different vendors produce them.

How Wireless Works

Wireless technologies employ radio waves and/or microwaves to maintain wireless communication channels between computers. While many technical details behind wireless protocols like Wi-Fi often aren't important to understand, knowing the basics about Wi-Fi can be very helpful when configuring a network and troubleshooting problems. The wireless tech we know today had its origins in scientific research going back several decades. Nikola Tesla pioneered wireless electric lighting and power transmission, for example—areas that continue to be an active area of study today for such usages as wireless charging.

IEEE WIRELESS STANDARDS

802.11 Wireless Standards

802.11 represents the IEEE designation for wireless networking. Several wireless networking specifications exist under the 802.11 banner. The Network+ objectives focus on 802.11, 802.11a, 802.11b, 802.11g, and 802.11n. All these standards use the Ethernet protocol and the CSMA/CA access method. The 802.11 wireless standards can differ in terms of speed, transmission ranges, and frequency used, but in terms of actual implementation they are similar. All standards can use either an infrastructure or ad hoc network design, and each can use the same security protocols.

1. **IEEE 802.11:** There were actually two variations on the initial 802.11 wireless standard. Both offered 1 or 2Mbps transmission speeds and the same RF of 2.4GHz. The difference between the two was in how data traveled through the RF media. One used FHSS, and the other used DSSS. The original 802.11 standards are far too slow for modern networking needs and are now no longer deployed.

2. **IEEE 802.11a:** In terms of speed, the 802.11a standard was far ahead of the original 802.11 standards. 802.11a specified speeds of up to 54Mbps in the 5GHz band, but most commonly, communication takes place at 6Mbps, 12Mbps, or 24Mbps. 802.11a is incompatible with the 802.11b and 802.11g wireless standards.
3. **IEEE 802.11b:** The 802.11b standard provides for a maximum transmission speed of 11Mbps. However, devices are designed to be backward-compatible with previous 802.11 standards that provided for speeds of 1, 2, and 5.5Mbps. 802.11b uses a 2.4GHz RF range and is compatible with 802.11g.
4. **IEEE 802.11g:** 802.11g is a popular wireless standard today. 802.11g offers wireless transmission over distances of 150 feet and speeds up to 54Mbps compared with the 11Mbps of the 802.11b standard. Like 802.11b, 802.11g operates in the 2.4GHz range and therefore is compatible with it.
5. **IEEE 802.11n:** The newest of the wireless standards listed in the Network+ objectives is 802.11n. The goal of the 802.11n standard is to significantly increase throughput in both the 2.4GHz and the 5GHz frequency range. The baseline goal of the standard was to reach speeds of 100Mbps, but given the right conditions, it is estimated that the 802.11n speeds might reach a staggering 600Mbps. In practical operation, 802.11n speeds will be much slower.

802.11 Wireless Standards

IEEE Standard	Frequency/ Medium	Speed	Topology	Transmission Range	Access Method
802.11	2.4GHz RF	1 to 2Mbps	Ad hoc/infrastructure	20 feet indoors.	CSMA/CA
802.11a	5GHz	Up to 54Mbps	Ad hoc/infrastructure	25 to 75 feet indoors; range can be affected	CSMA/CA

				by building materials.	
802.11b	2.4GHz	Up to 11Mbps	Ad hoc/infrastructure	Up to 150 feet indoors; range can be affected by building materials.	CSMA/CA
802.11g	2.4GHz	Up to 54Mbps	Ad hoc/infrastructure	Up to 150 feet indoors; range can be affected by building materials.	CSMA/CA
802.11n	2.4GHz/5GHz	Up to 600Mbps	Ad hoc/infrastructure	175+ feet indoors; range can be affected by building materials.	CSMA/CA

FHSS, DSSS, OFDM, and 802.11 Standards

The original 802.11 standard had two variations, both offering the same speeds but differing in the RF spread spectrum used. One of the 802.11 standards used FHSS. This 802.11 variant used the 2.4GHz radio frequency band and operated at a 1 or 2Mbps data rate. Since this original standard, wireless implementations have favored DSSS. The second 802.11 variation used DSSS and specified a 2Mbps peak data rate with optional fallback to 1Mbps in very noisy environments. 802.11, 802.11b, and 802.11g use DSSS. This means that the underlying modulation scheme is similar between each standard, allowing all DSSS systems to coexist with 2, 11, and 54Mbps 802.11 standards. As a comparison, it is like the migration from the older 10Mbps Ethernet networking to the more commonly implemented 100Mbps standard. The speed was different, but the underlying technologies were similar, allowing for an easier upgrade.

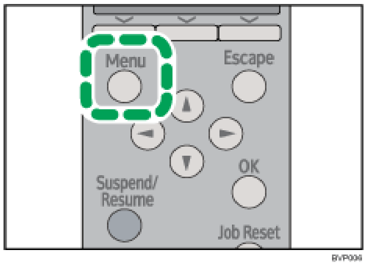
Comparison of IEEE 802.11 Standards

IEEE Standard	RF Used	Spread Spectrum	Data Rate (in Mbps)
802.11	2.4GHz	DSSS	1 or 2
802.11	2.4GHz	FHSS	1 or 2
802.11a	5GHz	OFDM	54
802.11b	2.4GHz	DSSS	11
802.11g	2.4Ghz	DSSS	54
802.11n	2.4/5GHz	OFDM	600 (theoretical)

Wireless LAN Configuration

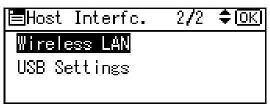
Ethernet and Wireless LAN cannot be used at the same time. To use wireless LAN, set as follows using the control panel: press the [Menu] key, select [Host Interface], [Network], [LAN Type], and then select [Wireless LAN]. In addition, you must set the IP address, subnet mask, gateway address, DHCP, Frame Type (NW), and active protocol as explained in "Ethernet Configuration". The following table shows the control panel settings and their defaults. These items appear in the [Host Interface] menu.

1. **Press the [Menu] key.**

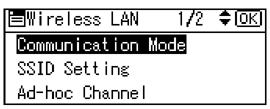


2. **Press the [▼] or [▲] key to select [Host Interface], and then press the [OK] key.**

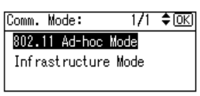
3. **Press the [▼] or [▲] key to select [Wireless LAN], and then press the [OK] key.**



4. **Press the [▼] or [▲] key to select [Communication Mode], and then press the [OK] key.**

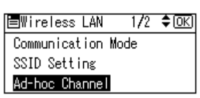


5. **Press the [▼] or [▲] key to select the transmission mode of the wireless LAN, and then press the [OK] key.**

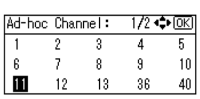


- To communicate wirelessly with a device that does not require an SSID to be set, select [802.11 Ad-hoc Mode].
- The transmission mode of the wireless LAN can also be set using Web Image Monitor.

6. **If [802.11 Ad-hoc Mode] is selected for [Communication Mode], press the [▼] or [▲] key to select [Ad-hoc Channel], and then press the [OK] key.**



7. **Press the scroll keys to select the Ad-hoc channel, and then press the [OK] key.**



Select an Ad-hoc channel according to the IEEE 802.11 standard you are using.

When using IEEE 802.11 b/g:

220-240V Channels 1 to 13

120V Channels 1 to 11

REVIEW QUESTIONS

Name : ________________________________ Score :_________________
Section: ________________________________ Date : _________________

I. Write your answers on the space provided after the number.

_________1. There is a portion of an old network in your building that has backbone cable marked as RG-8. When you measure the impedance of the cable, what should be the result showing that the cable is in a good working condition?

A. 50 ohms

B. 35 ohms

C. 22.5 ohms

D. 18 ohms

_________2. Which of the following transmissions uses several transmission channels on a single communications medium?

A. Bursty

B. Baseband

C. Multicapacity

D. Broadband

_________3. You are installing UTP cable in a building to ensure up to 1Gbps to the desktop. Which of the following cable types provide assured handling of 1 Gbps? (Choose all that apply.)

A. Category 3

B. Category 5

C. Category 5e

D. Category 6a

_________4. Gigabit Ethernet uses which of the following? (Choose all that apply.)

A. Headend termination

B. Fiber-optic, twisted-pair, and coax cable options

C. CSMA/CD

D. Application Layer padding

_________5. What type of communications cable is used by 10GBaseF?

A. fiber-optic

B. thick coax

C. twisted pair

D. thin coax

_________6. You have a new computer and a newly released NIC, but when you try to connect to your school's network, your computer frequently loses the network connection. Which of the following might be the problem? (Choose all that apply.)

A. You NIC in the PCI slot, which does not offer fast enough communications with the computer.

B. Your school uses a ring network, and this is a common problem with these networks.

C. You forgot to disable the firmware chip for the PCI slot, which interferes with the firmware chip in the NIC.

D. The driver for the new NIC has some software bugs and you should download a more recent driver in which the bugs are fixed.

_________7. Hybrid fiber/coax cables are typically used for which type of installation?

A. Cable to the desktop

B. The NIC's cable connection to a computer's motherboard

C. Networks that use token timing

D. Cable TV infrastructure

_________8. You are consulting for a real estate holding company that is considering the purchase of an office building in which there is 100BaseT cable and network equipment installed to each of the offices. You report to the real estate holding company that the speed of transmissions to the network sockets in each office is which of the following?

A. 100 Kbps

B. 100 Mbps

C. 100 Gbps

D. 100 Tbps

_________9. Which type of fiber-optic cable is best for long-distance communications of up to 40 miles or even more?

A. Single-mode

B. Multimode

C. Thicknet

D. Distance-modulated

_________10. Which of the following are the roles played by the MAC controller unit combined with the firmware of a NIC? (Choose all that apply.)

A. Correctly encapsulating the source and destination address information in packets and d. setting CRC error control information

B. Detecting whether communications are duplex or nonduplex

C. Adjusting the signal tone of packets

D. Setting CRC error control information

_________11. You work as a security consultant for federal government agencies, such as the FBI, and you need to make sure that your work computer running Windows 7 is secured. One step that you can take is to do which of the following in relation to your computer's NIC driver?

A. Make sure you use a 32-bit driver.

B. Use a fast-wireless connection so that it is harder for intruders to capture data from your computer.

C. Confirm proper driver signing for the NIC driver.

D. Use only a NIC integrated on the motherboard because this type of NIC is from a secure manufacturer.

_________12. When you install a 100BaseX twisted-pair cable segment, what should be the length of cable in order to allow for wiring in network devices and also ensure you are within the wiring specifications?

A. 330 meters

B. 225 meters

C. 124 meters

D. 90 meters

_________13. Step index applies to which type of cable?

A. Graded serial single-mode fiber-optic cable

B. Multimode fiber-optic cable

C. Thicknet

D. Thinner

_________14. One of the users that you support in your company has purchased a new laptop. He purchased the computer at a computer outlet store, and a salesperson manually configured the NIC for half-duplex communications. When he attaches the laptop to the network, and it does not properly communicate. What might you do first to fix the problem?

A. Set the NIC for 805.ab communications, which is required to use half duplex.

B. Purchase a transceiver to use between the NIC and the network cable.

C. Reset the computer to use full-duplex communications, because that is used by connecting devices on the network.

D. Recommend that he take the laptop back to the store, because modern networks no longer use duplex communications.

_________15. of the following are the types of connectors you might find used with fiber-optic cable? (Choose all that apply.)

A. Fiber-tip BNC

B. Subscriber connector (SC)

C. Lucent Connector (LC)

D. Mechanical Transfer Registered Jack (MT-RJ)

_________16. Which of the following versions of Fast Ethernet use CSMA/CD? (Choose all that apply.)

A. 100BaseFX

B. 100BaseT4

C. 100BaseT2

D. 100BaseTX

_________17. 10 Gigabit Ethernet operates only at which of the following?

A. full duplex

B. 142 MHz

C. singleplex

D. 10,5482 Gbps

_________18. Which of the following cable types are used with 100GBaseSR4? (Choose all that apply.)

A. OM3

B. OM4

C. UTP Cat 7r

D. UTP Cat 9SC

_________19. The president of your company has replaced her two-year-old Microsoft computer with a newer one and wants you, as the network consultant, to set up her old computer for her administrative assistant. You realize that the NIC in the president's old computer, which is in the EISA slot, is damaged, and so you decide to replace the NIC. What type of slot for a NIC in the computer should you look for before ordering a new NIC?

A. VL-bus

B. NuBus

C. SBus

D. PCI

__________20. The IT manager in your company wants you to be able to build twisted-pair cable so there is an alternative if the company unexpectedly runs out of prebuilt cable. Plus the IT manager believes this will be good knowledge for cable troubleshooting. What cable connectors should you order to go with the twisted-pair cable?

A. RG-58

B. RJ-11

C. RJ-45

D. RG-58 A/U

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